Asymptotic notation

Outline

Introduction

Θ, $\mathcal{O}$ and Ω notations

o and ω notations

Limits and asymptotic notations

Relational properties of asymptotic notations

Asymptotic notation in the analysis of algorithms

Prerequisites

- The functions of a natural variable are the **sequences** of the course *Mathematical Analysis 1* (fabiofurini.github.io/mathematical-analysis-1/, Italian version fabiofurini.github.io/analisi-matematica-1/)
- Their limits, and the theorems used to compute them, are not explained again in this chapter
- We use the definition of limit and the rules to compute limits of the part *Limits of sequences* of that course

Introduction

Introduction: comparing algorithms

- The number of steps executed by an algorithm is a function of a natural number: the **size of its input**
- Two algorithms that solve the same problem are compared by looking at how fast these functions grow when the input grows
- We neglect constant factors and the values on a few small inputs
- In this chapter we give the precise language used to express these comparisons: the **asymptotic notations** Θ, $\mathcal{O}$, Ω, o and ω

Plan of the chapter (1/2)

- We define the notations Θ, $\mathcal{O}$ and Ω, we interpret them graphically and we prove that Θ is the same as $\mathcal{O}$ and Ω together; we also study the sum and the maximum of two functions
- We define the notations o and ω, which express bounds that are **not** asymptotically tight
- We link the five notations to the limit of the ratio $\frac{f(n)}{g(n)}$: when the limit exists, its value decides which of the five notations hold (the **limit criterion**); when it does not exist, we use the definitions

Plan of the chapter (2/2)

- We prove the relational properties of the notations (transitivity, reflexivity, symmetry and transpose symmetry): they make the asymptotic comparison of two functions similar to the comparison of two real numbers, with one important difference
- We see how the notation is used in the analysis of algorithms of the chapters «Input size» and «Running time»
- Every constant (c, c_1, c_2, n_1) used in the examples of the chapter has been checked numerically with a script, for all the values of n from the threshold n_1 up to at least $2 \cdot 10^5$

Terminology

Throughout the chapter we use the following terminology.

Definition 1: asymptotically non-negative and asymptotically positive functions

*A function f of a natural variable is **asymptotically non-negative** if there exists $\tilde{n} \in \mathbb{N}$ such that $f(n) \geq 0$ for all $n \geq \tilde{n}$.*

*It is **asymptotically positive** if there exists $\tilde{n} \in \mathbb{N}$ such that $f(n) > 0$ for all $n \geq \tilde{n}$.*

Summary

- The number of steps of an algorithm is a function of a natural number, the size of the input
- Algorithms are compared through the **growth** of these functions, neglecting constant factors and small inputs
- The functions we compare are **asymptotically non-negative** (or **asymptotically positive**): non-negative (positive) from some $\tilde{n}$ on

Θ, $\mathcal{O}$ and Ω notations

Introduction: Θ, $\mathcal{O}$ and Ω notations

- We define the three basic notations: a function bounded from above and from below (Θ), from above ($\mathcal{O}$), from below (Ω) by another one, up to constants
- We learn how to prove such statements, by determining the constants, and how to disprove them, by contradiction
- We prove that Θ is the same as $\mathcal{O}$ and Ω together, and we study the sum and the maximum of two functions
- These notations are the language of the whole analysis of algorithms

Definition 2: Θ-notation

Given two functions f and g of a natural variable, we write $f(n) = \Theta\big(g(n)\big)$ if there exist $c_1, c_2 \in \mathbb{R}_{>0}$ and $n_1 \in \mathbb{N}$ such that

$$0 \leq c_1\, g(n) \leq f(n) \leq c_2\, g(n), \qquad \forall n \geq n_1. \tag{1}$$

Asymptotic tight bound

- If $f(n) = \Theta\big(g(n)\big)$, then, for sufficiently large values of n, the function f is bounded from above and from below by the function g scaled by two positive constants
- In this case we say that the function g is an **asymptotic tight bound** for the function f
- We write $f(n) = \Theta(1)$ to mean that the values $f(n)$ are, for sufficiently large n, bounded from above and from below by two positive constants (here g is the constant function $g(n) = 1$)

Definition 3: $\mathcal{O}$-notation

Given two functions f and g of a natural variable, we write $f(n) = \mathcal{O}\big(g(n)\big)$ if there exist $c \in \mathbb{R}_{>0}$ and $n_1 \in \mathbb{N}$ such that

$$0 \leq f(n) \leq c\, g(n), \qquad \forall n \geq n_1. \tag{2}$$

- If $f(n) = \mathcal{O}\big(g(n)\big)$, then, for sufficiently large values of n, the function f is bounded from above by the function g scaled by a positive constant
- In this case we say that the function g is an **asymptotic upper bound** for the function f

Definition 4: Ω-notation

Given two functions f and g of a natural variable, we write $f(n) = \Omega\big(g(n)\big)$ if there exist $c \in \mathbb{R}_{>0}$ and $n_1 \in \mathbb{N}$ such that

$$0 \leq c\, g(n) \leq f(n), \qquad \forall n \geq n_1. \tag{3}$$

- If $f(n) = \Omega\big(g(n)\big)$, then, for sufficiently large values of n, the function f is bounded from below by the function g scaled by a positive constant
- In this case we say that the function g is an **asymptotic lower bound** for the function f

The two functions are asymptotically non-negative

- If $f(n) = \Theta\big(g(n)\big)$, $f(n) = \mathcal{O}\big(g(n)\big)$ or $f(n) = \Omega\big(g(n)\big)$ then, since the constants c_1 and c are positive, we have

$$g(n) \geq 0 \quad \text{and} \quad f(n) \geq 0, \qquad \forall n \geq n_1$$

- For $\mathcal{O}$, the inequality $0 \leq f(n) \leq c\, g(n)$ gives $g(n) \geq 0$ as well

- Therefore, the two functions f and g are **asymptotically non-negative**

How to prove and disprove these statements

- To prove one of these statements we **determine** the constants (c, or c_1 and c_2) and the threshold n_1
- In general, the threshold n_1 is not the smallest possible one and the constants are not unique: any larger threshold, any smaller c_1 and any larger c_2 (or c in the definition of $\mathcal{O}$) also work
- To prove that a statement is false, we proceed by **contradiction**: we suppose that the constants exist and we derive an inequality that cannot hold for arbitrarily large n

We now see two examples of asymptotic tight bounds: in each of them we determine the constants c_1, c_2 and the threshold n_1.

Example: asymptotic tight bound (1/3)

- To prove that

$$-3\,n + \frac{1}{2}\,n^2 = \Theta(n^2)$$

we determine $c_1, c_2 \in \mathbb{R}_{>0}$ and $n_1 \in \mathbb{N}$ such that

$$0 \leq c_1\,n^2 \leq -3\,n + \frac{1}{2}\,n^2 \leq c_2\,n^2, \qquad \forall n \geq n_1.$$

- With $n \geq 1$, dividing by n^2 and since $c_1 > 0$, this is equivalent to

$$0 < c_1 \leq \frac{1}{2} - \frac{3}{n} \leq c_2, \qquad \forall n \geq n_1.$$

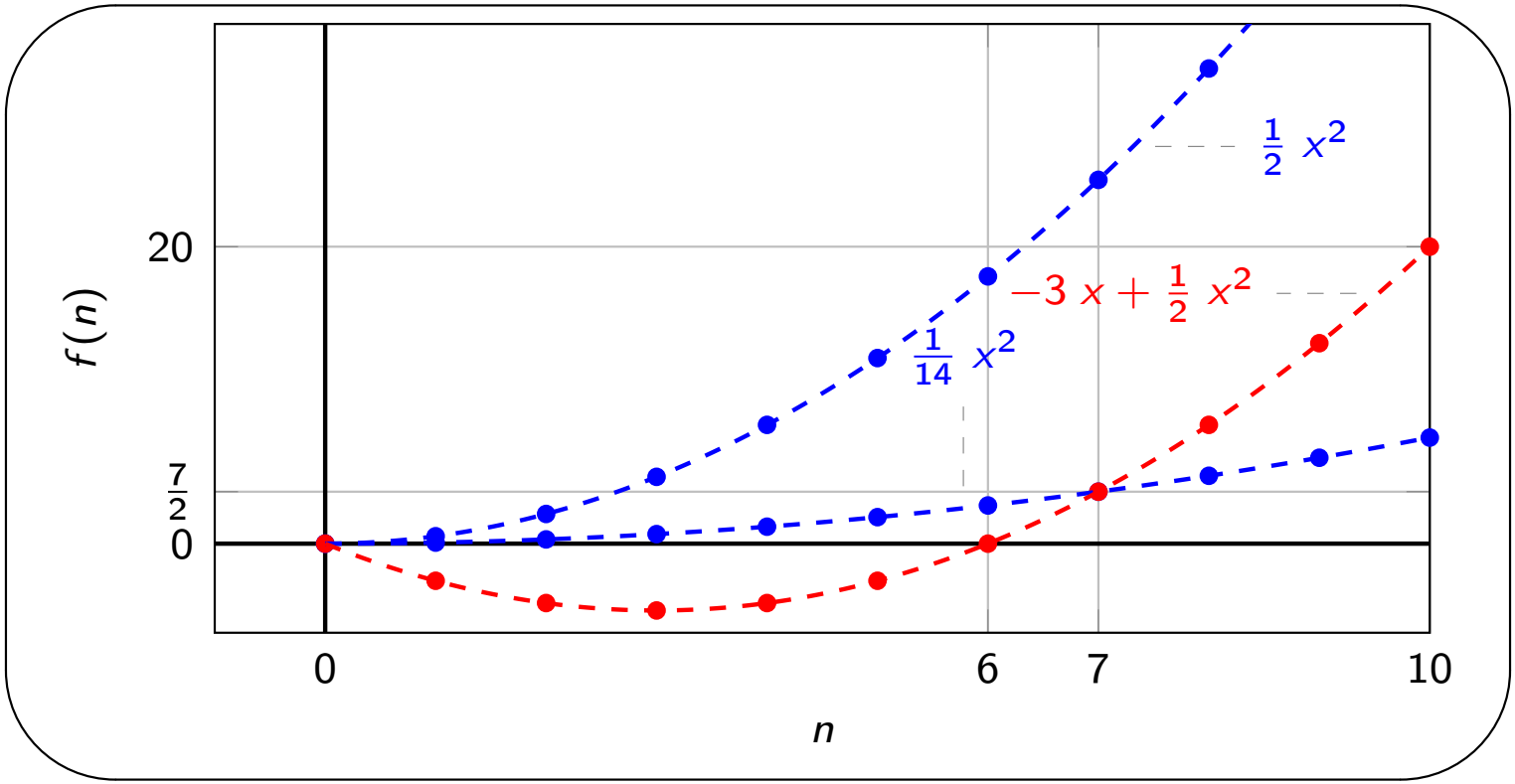
Example: asymptotic tight bound (2/3)

- The middle term increases with n and it is always smaller than $\frac{1}{2}$
- Thus, by choosing for example $c_1 = \frac{1}{14}$, $c_2 = \frac{1}{2}$ and $n_1 = 7$ (we selected $\frac{1}{14}$ since $\frac{1}{14} = \frac{1}{2} - \frac{3}{7}$), we have

$$0 \leq \frac{1}{14}\, n^2 \leq -3\, n + \frac{1}{2}\, n^2 \leq \frac{1}{2}\, n^2, \qquad \forall n \geq 7.$$

- For $n = 6$ the lower bound fails, since $-18 + 18 = 0 < \frac{36}{14}$

Example: asymptotic tight bound (3/3)



The previous proof also proves that $-3\,n+\frac{1}{2}\,n^2=\mathcal{O}(n^2)$ and $-3\,n+\frac{1}{2}\,n^2=\Omega(n^2)$.

Example: asymptotic tight bound (1/3)

- To prove that
$$n^2 \, (2 - \sin n) = \Theta(n^2)$$
we determine $c_1, c_2 \in \mathbb{R}_{>0}$ and $n_1 \in \mathbb{N}$ such that
$$0 \leq c_1 \, n^2 \leq n^2 \, (2 - \sin n) \leq c_2 \, n^2, \qquad \forall n \geq n_1.$$
- With $n \geq 1$, dividing by n^2 and since $c_1 > 0$, this is equivalent to
$$0 < c_1 \leq 2 - \sin n \leq c_2, \qquad \forall n \geq n_1.$$

Example: asymptotic tight bound (2/3)

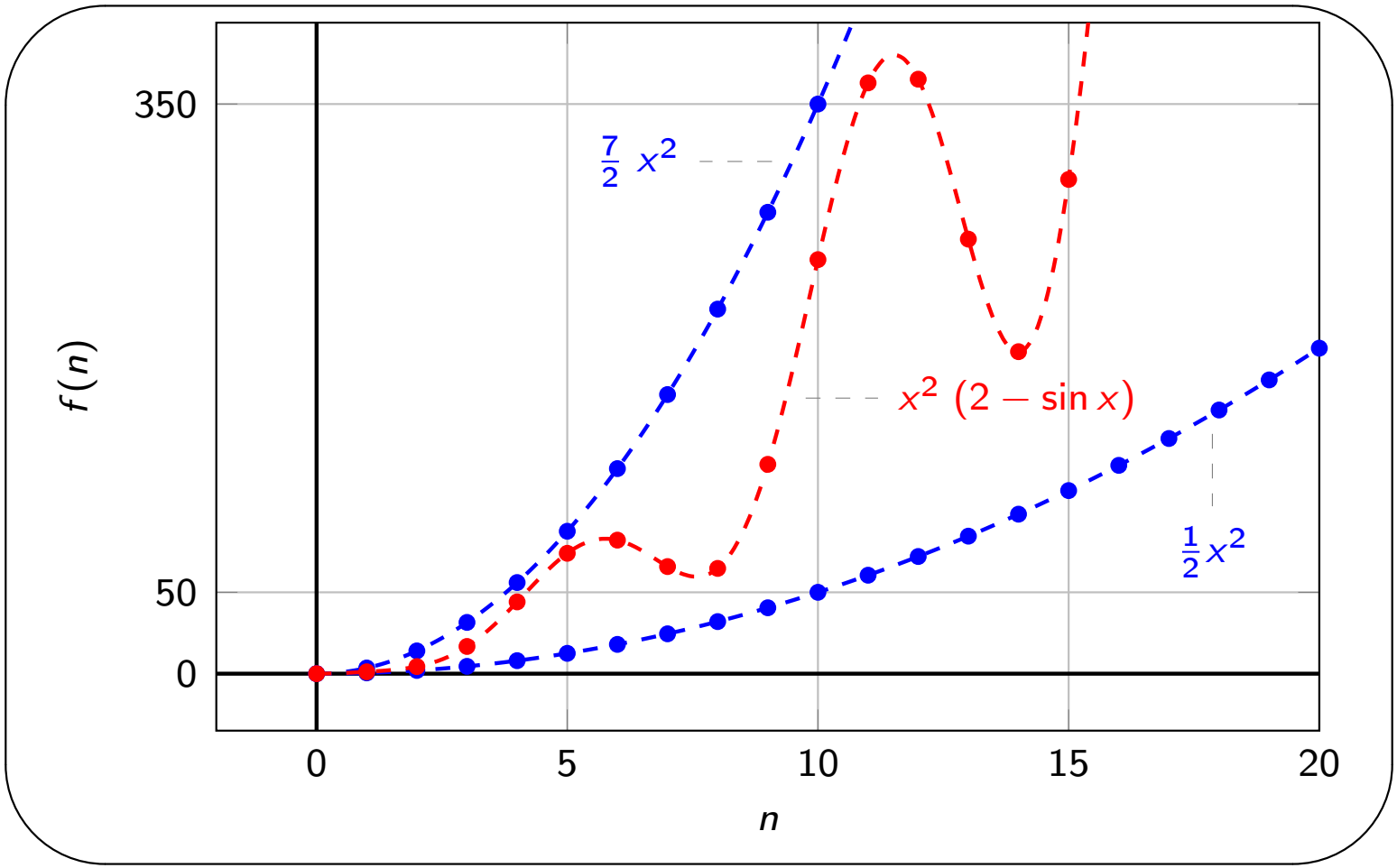
- Since $-1 \leq \sin n \leq 1$, we have $1 \leq 2 - \sin n \leq 3$
- Thus, by choosing for example $c_1 = \frac{1}{2}$, $c_2 = \frac{7}{2}$ and $n_1 = 1$, we have

$$0 \leq \frac{1}{2}\, n^2 \leq n^2\,(2 - \sin n) \leq \frac{7}{2}\, n^2, \qquad \forall n \geq 1.$$

- The previous proof also proves that

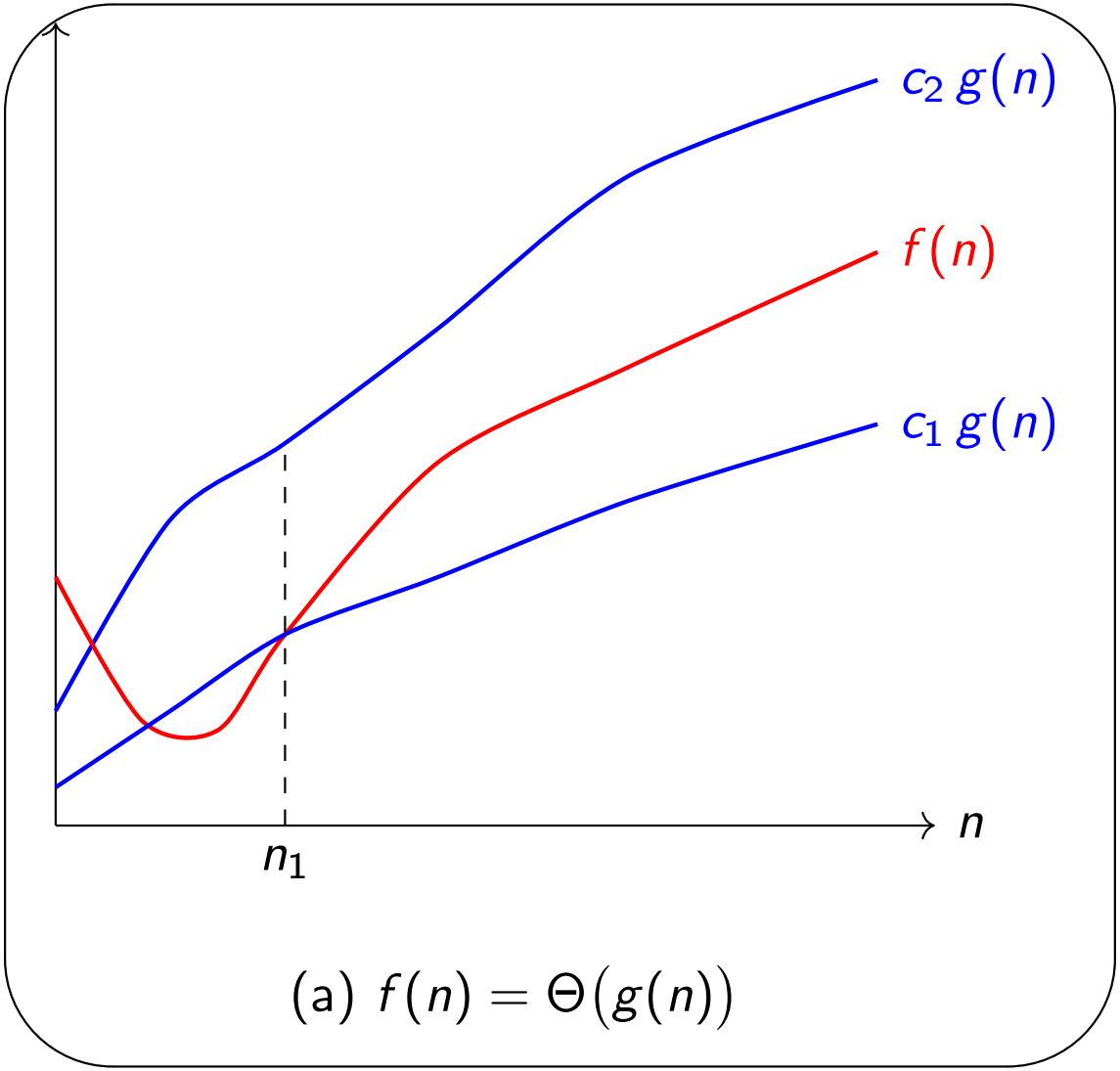
$$n^2\,(2 - \sin n) = \mathcal{O}(n^2) \qquad \text{and} \qquad n^2\,(2 - \sin n) = \Omega(n^2).$$

Example: asymptotic tight bound (3/3)



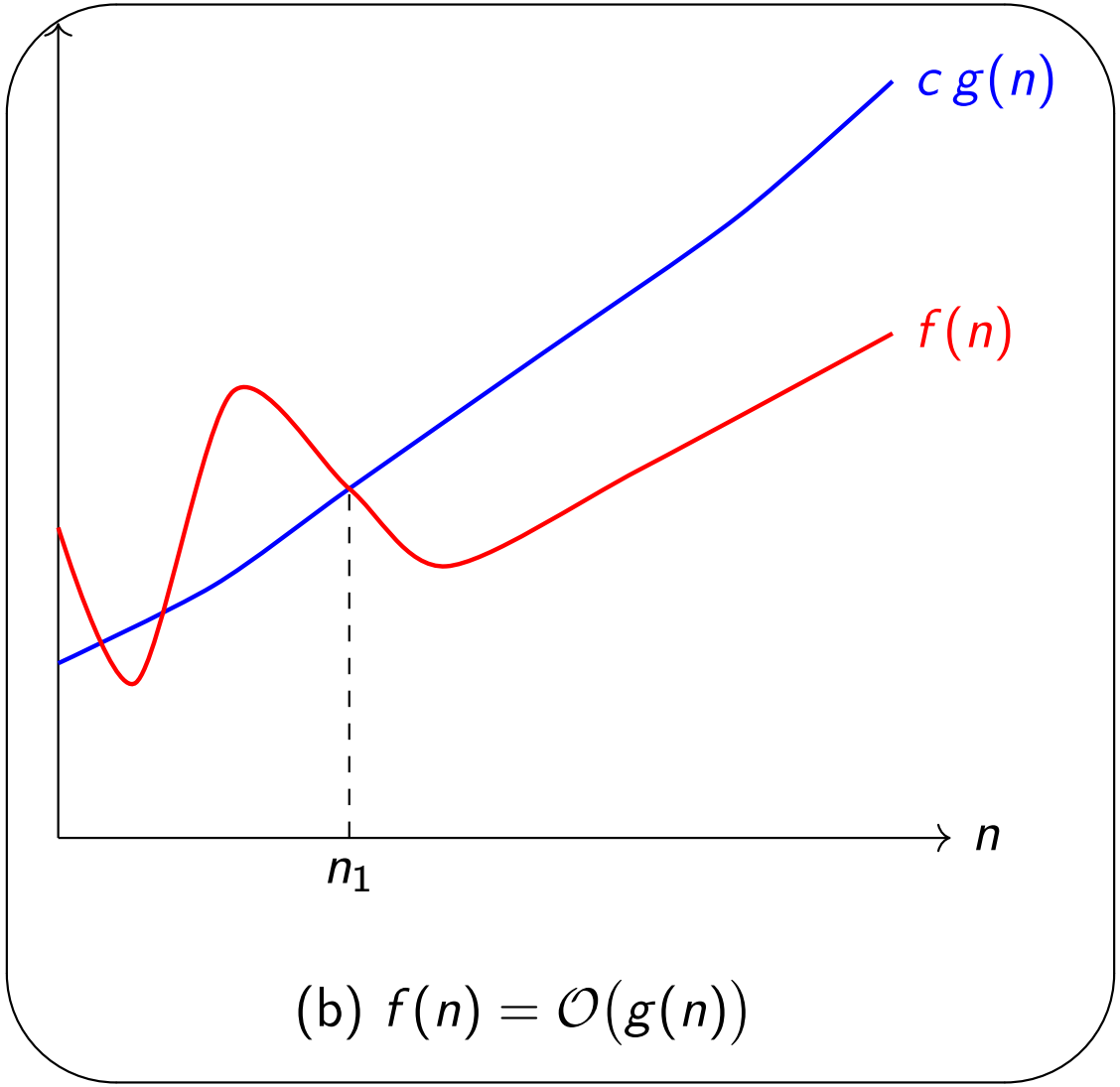
The next three figures give a graphical interpretation of the three notations Θ, $\mathcal{O}$ and Ω.

Graphical interpretation (1/4)



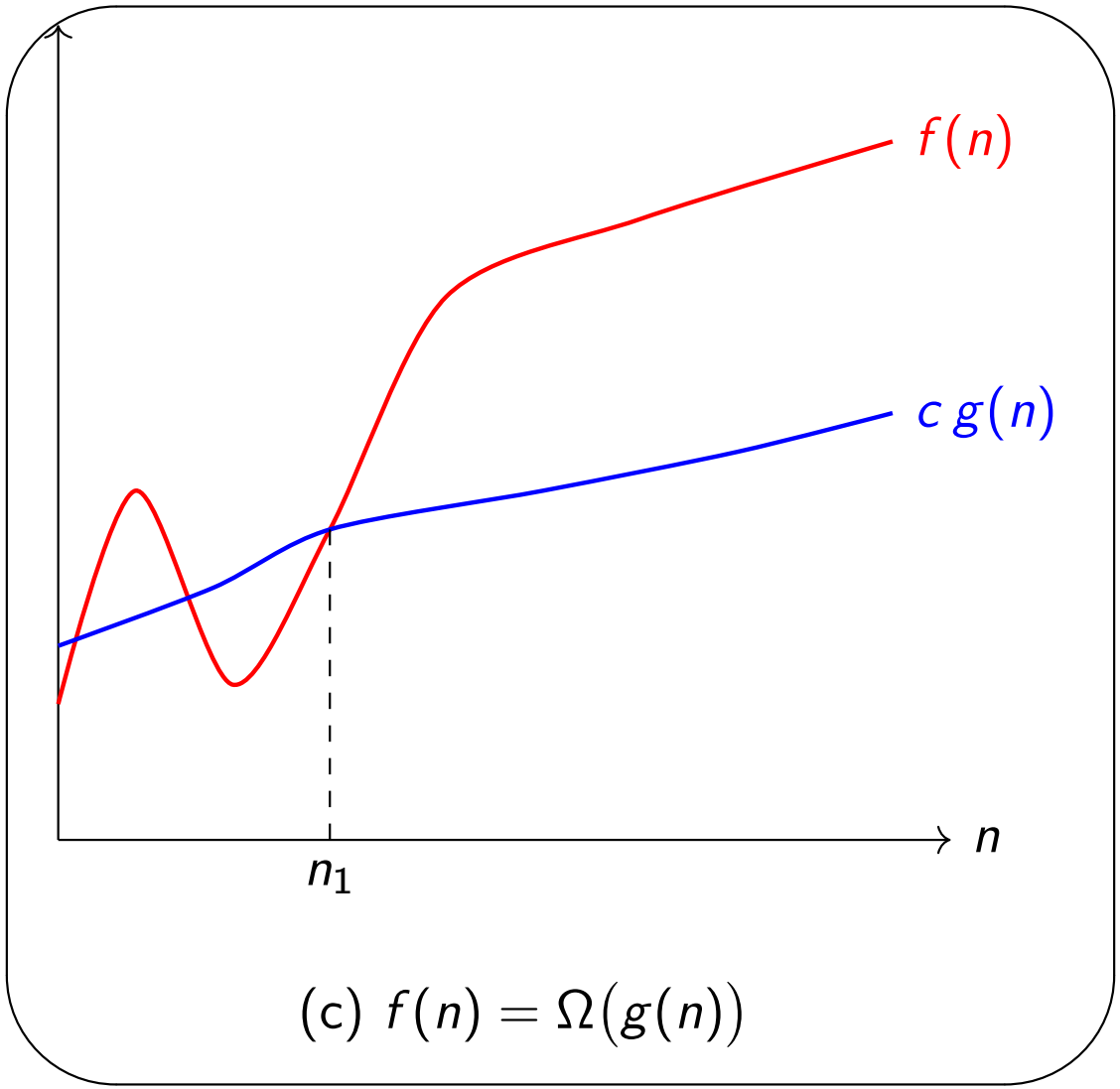
(a) $f(n) = \Theta\big(g(n)\big)$

Graphical interpretation (2/4)



(b) $f(n) = \mathcal{O}\big(g(n)\big)$

Graphical interpretation (3/4)



(c) $f(n) = \Omega\big(g(n)\big)$

Graphical interpretation (4/4)

- In (a), from n_1 on, the values of f lie between $c_1\, g(n)$ and $c_2\, g(n)$
- In (b) they lie below $c\, g(n)$
- In (c) they lie above $c\, g(n)$
- Before n_1 nothing is required
- In the figures, n_1 is the smallest possible threshold, but any larger value would also work
- The figures also illustrate one possible choice of the constants c_1, c_2 and c, but other valid choices exist

How are the three notations related?
We prove that Θ is exactly $\mathcal{O}$ and Ω together.

Observation 1: Θ is the same as $\mathcal{O}$ and Ω

Given two functions f and g of a natural variable, we have

$$f(n) = \Theta\big(g(n)\big) \iff f(n) = \mathcal{O}\big(g(n)\big) \text{ and } f(n) = \Omega\big(g(n)\big).$$

In particular,

$$f(n) = \Theta\big(g(n)\big) \implies f(n) = \mathcal{O}\big(g(n)\big),$$
$$f(n) = \Theta\big(g(n)\big) \implies f(n) = \Omega\big(g(n)\big).$$

Proof

- If $f(n) = \Theta\big(g(n)\big)$, the two inequalities

$$0 \leq f(n) \leq c_2\, g(n) \qquad \text{and} \qquad 0 \leq c_1\, g(n) \leq f(n)$$

  contained in (1) give $f(n) = \mathcal{O}\big(g(n)\big)$ (with $c = c_2$) and $f(n) = \Omega\big(g(n)\big)$ (with $c = c_1$), with the same threshold n_1
- Conversely, let $0 \leq f(n) \leq c'\, g(n)$ for all $n \geq n_1'$, and $0 \leq c''\, g(n) \leq f(n)$ for all $n \geq n_1''$
- By choosing $c_1 = c''$, $c_2 = c'$ and $n_1 = \max\{n_1', n_1''\}$, we obtain

$$0 \leq c_1\, g(n) \leq f(n) \leq c_2\, g(n), \qquad \forall n \geq n_1,$$

  that is, $f(n) = \Theta\big(g(n)\big)$



Necessary and sufficient conditions (1/2)

- It follows that «$f(n) = \Theta\big(g(n)\big)$» is a sufficient condition for «$f(n) = \mathcal{O}\big(g(n)\big)$», and «$f(n) = \mathcal{O}\big(g(n)\big)$» is a necessary condition for «$f(n) = \Theta\big(g(n)\big)$»
- By contraposition we also have

$$f(n) \neq \mathcal{O}\big(g(n)\big) \implies f(n) \neq \Theta\big(g(n)\big).$$

- The implication does not work in the other direction (see the next two examples), that is,

$$f(n) = \mathcal{O}\big(g(n)\big) \not\Rightarrow f(n) = \Theta\big(g(n)\big).$$

Necessary and sufficient conditions (2/2)

- In the same way, «$f(n) = \Theta\big(g(n)\big)$» is a sufficient condition for «$f(n) = \Omega\big(g(n)\big)$», and «$f(n) = \Omega\big(g(n)\big)$» is a necessary condition for «$f(n) = \Theta\big(g(n)\big)$»
- By contraposition we also have

$$f(n) \neq \Omega\big(g(n)\big) \implies f(n) \neq \Theta\big(g(n)\big),$$

- and the implication does not work in the other direction (see the third example below):

$$f(n) = \Omega\big(g(n)\big) \not\Rightarrow f(n) = \Theta\big(g(n)\big).$$

Example: upper bound that is not tight (1/4)

- To prove that

$$3\,n^2 = \mathcal{O}(n^3)$$

we determine $c \in \mathbb{R}_{>0}$ and $n_1 \in \mathbb{N}$ such that

$$0 \leq 3\,n^2 \leq c\,n^3, \qquad \forall n \geq n_1.$$

- With $n \geq 1$, dividing by n^3, this is equivalent to

$$0 \leq \frac{3}{n} \leq c, \qquad \forall n \geq n_1.$$

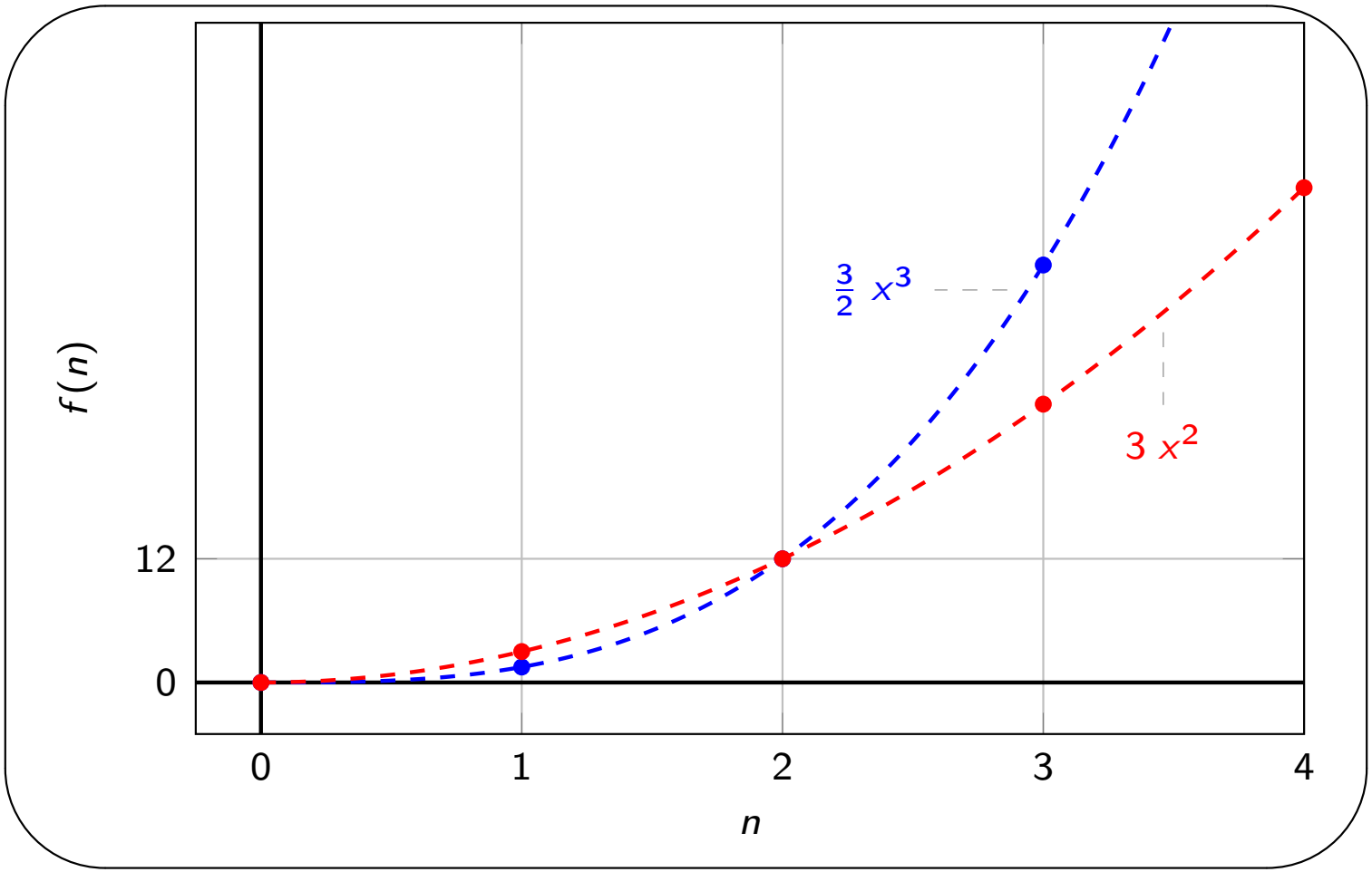
Example: upper bound that is not tight (2/4)

- Thus, by choosing for example $c = \frac{3}{2}$ and $n_1 = 2$, we have

$$0 \leq 3\, n^2 \leq \frac{3}{2}\, n^3, \qquad \forall n \geq 2.$$

- For $n = 1$ the inequality fails, since $3 > \frac{3}{2}$

Example: upper bound that is not tight (3/4)



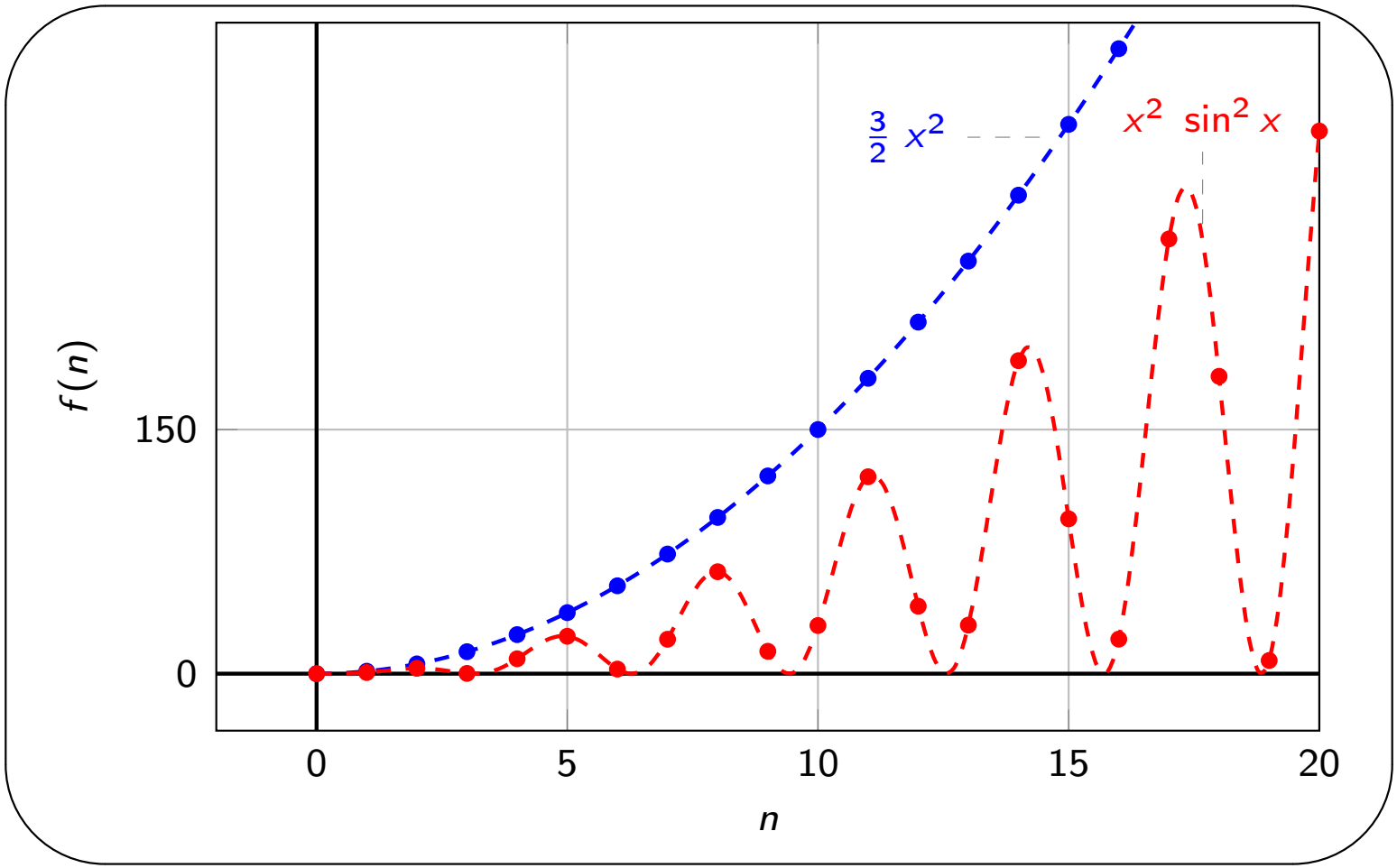
Example: upper bound that is not tight (4/4)

- To prove that
$$3\,n^2 \neq \Theta(n^3)$$
we proceed by contradiction and we suppose that there exist $c_1 \in \mathbb{R}_{>0}$ and $n_1 \in \mathbb{N}$ such that
$$0 \leq c_1\,n^3 \leq 3\,n^2, \qquad \forall n \geq n_1.$$
- With $n \geq 1$, dividing by n^3 and since $c_1 > 0$, this is equivalent to
$$0 < c_1 \leq \frac{3}{n}, \qquad \forall n \geq n_1,$$
which cannot hold for arbitrarily large n, since c_1 is a positive constant (it fails for every $n > \frac{3}{c_1}$)

Example: upper bound that is not tight (1/4)

- To prove that
$$n^2 \sin^2 n = \mathcal{O}(n^2)$$
we determine $c \in \mathbb{R}_{>0}$ and $n_1 \in \mathbb{N}$ such that
$$0 \leq n^2 \sin^2 n \leq c\, n^2, \qquad \forall n \geq n_1.$$
- With $n \geq 1$, dividing by n^2, this is equivalent to
$$0 \leq \sin^2 n \leq c, \qquad \forall n \geq n_1.$$
- Since $0 \leq \sin^2 n \leq 1$, by choosing for example $c = \frac{3}{2}$ and $n_1 = 1$ we have
$$0 \leq n^2 \sin^2 n \leq \frac{3}{2} n^2, \qquad \forall n \geq 1.$$

Example: upper bound that is not tight (2/4)



Example: upper bound that is not tight (3/4)

- To prove that
$$n^2 \sin^2 n \neq \Theta(n^2)$$
we proceed by contradiction and we suppose that there exist $c_1 \in \mathbb{R}_{>0}$ and $n_1 \in \mathbb{N}$ such that
$$0 \leq c_1 n^2 \leq n^2 \sin^2 n, \qquad \forall n \geq n_1.$$
- With $n \geq 1$, dividing by n^2 and since $c_1 > 0$, this is equivalent to
$$0 < c_1 \leq \sin^2 n, \qquad \forall n \geq n_1,$$
which cannot hold for arbitrarily large n, since c_1 is a positive constant (see the next slide)

Example: upper bound that is not tight (4/4)

Why $0 < c_1 \leq \sin^2 n$ cannot hold for all sufficiently large n:

- Since π is irrational, the values $n \bmod \pi$, $n \in \mathbb{N}$, are dense in $[0, \pi]$
- Therefore, for every $\varepsilon \in \mathbb{R}_{>0}$ there exist arbitrarily large integers n such that $|\sin n| < \varepsilon$
- By choosing $\varepsilon = \frac{\sqrt{c_1}}{2}$, we obtain

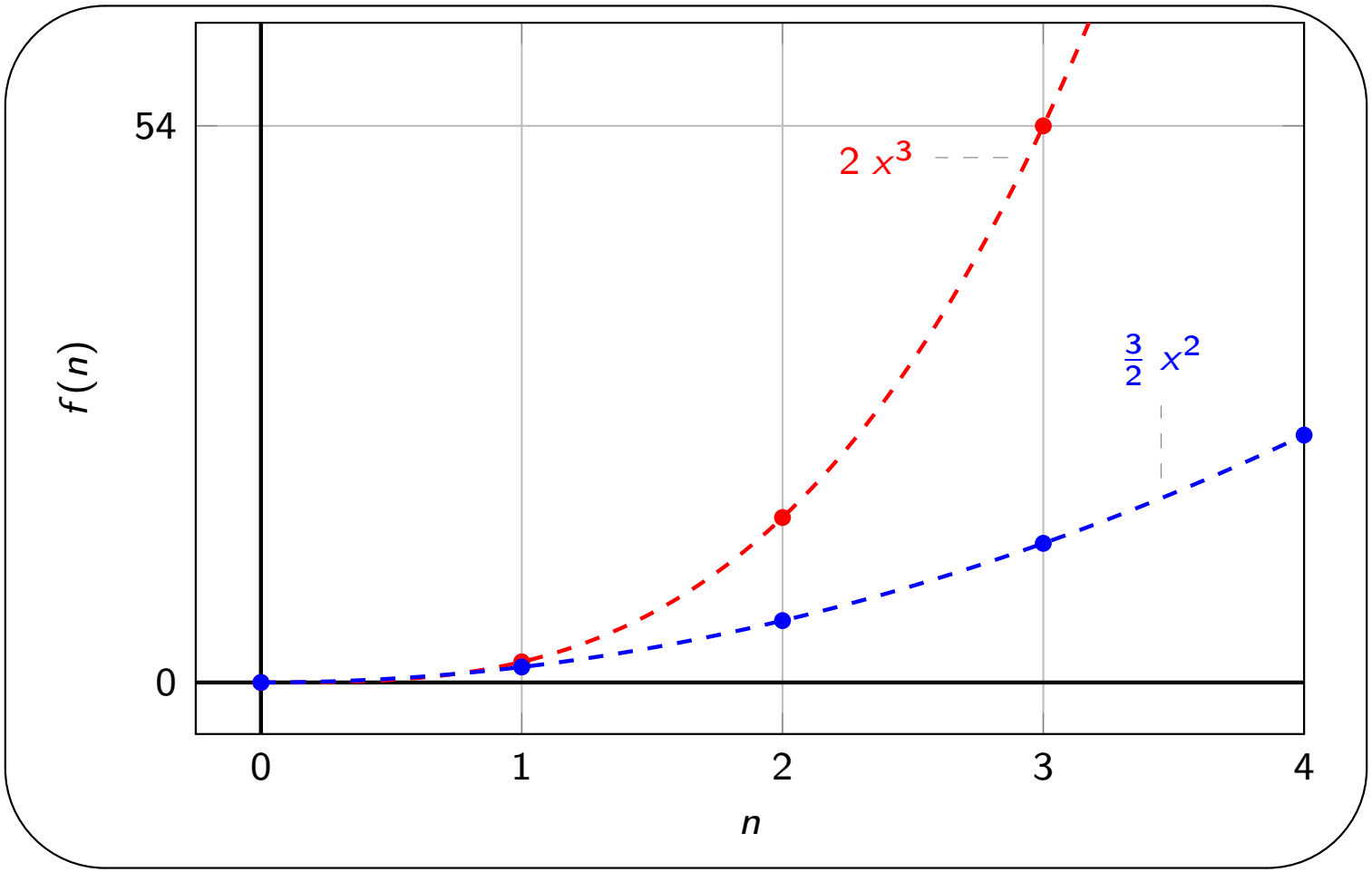
$$\sin^2 n < \frac{c_1}{4} < c_1,$$

which contradicts $c_1 \leq \sin^2 n$ for all sufficiently large n

Example: lower bound that is not tight (1/3)

- To prove that
$$2\,n^3 = \Omega(n^2)$$
we determine $c \in \mathbb{R}_{>0}$ and $n_1 \in \mathbb{N}$ such that
$$0 \leq c\,n^2 \leq 2\,n^3, \qquad \forall n \geq n_1.$$
- With $n \geq 1$, dividing by n^2 and since $c > 0$, this is equivalent to
$$0 < c \leq 2\,n, \qquad \forall n \geq n_1.$$
- Thus, by choosing for example $c = \frac{3}{2}$ and $n_1 = 1$, we have
$$0 \leq \frac{3}{2}\,n^2 \leq 2\,n^3, \qquad \forall n \geq 1.$$

Example: lower bound that is not tight (2/3)



Example: lower bound that is not tight (3/3)

- To prove that

$$2\,n^3 \neq \Theta(n^2)$$

we proceed by contradiction and we suppose that there exist $c_2 \in \mathbb{R}_{>0}$ and $n_1 \in \mathbb{N}$ such that

$$0 \leq 2\,n^3 \leq c_2\,n^2, \qquad \forall n \geq n_1.$$

- With $n \geq 1$, dividing by n^2, this is equivalent to

$$0 \leq 2\,n \leq c_2, \qquad \forall n \geq n_1,$$

which cannot hold for arbitrarily large n, since c_2 is a constant (it fails for every $n > \frac{c_2}{2}$)

We now study what happens
when we add two functions.

Observation 2: adding the bound to the function

Given two functions f and g of a natural variable, we have

$$f(n) = \Theta\big(g(n)\big) \implies f(n) + g(n) = \Theta\big(g(n)\big). \qquad (4)$$

Proof

- If $f(n) = \Theta\big(g(n)\big)$, there exist $c_1, c_2 \in \mathbb{R}_{>0}$ and $n_1 \in \mathbb{N}$ such that
$$0 \leq c_1\, g(n) \leq f(n) \leq c_2\, g(n), \qquad \forall n \geq n_1.$$
- Moreover, since $c_1 > 0$, we have $g(n) \geq 0$ for all $n \geq n_1$. Then, by adding $g(n)$, we get, for all $n \geq n_1$,
$$0 \leq \underbrace{c_1\, g(n) + g(n)}_{=\underbrace{(c_1+1)}_{>0}\, g(n)} \leq f(n) + g(n) \leq \underbrace{c_2\, g(n) + g(n)}_{=\underbrace{(c_2+1)}_{>0}\, g(n)}.$$
- Accordingly, $f(n) + g(n) = \Theta\big(g(n)\big)$, with the constants $c_1 + 1$ and $c_2 + 1$ and the same threshold n_1

□

Example: asymptotic tight bound of a sum (1/2)

- In an earlier example we proved that

$$-3\,n + \frac{1}{2}\,n^2 = \Theta(n^2)$$

with $c_1 = \frac{1}{14}$, $c_2 = \frac{1}{2}$ and $n_1 = 7$

- Then, according to (4), we have

$$-3\,n + \frac{1}{2}\,n^2 + n^2 = \Theta(n^2),$$

with the constants $c_1 + 1 = \frac{15}{14}$, $c_2 + 1 = \frac{3}{2}$ and $n_1 = 7$

Example: asymptotic tight bound of a sum (2/2)

- In an earlier example we proved that

$$n^2 (2 - \sin n) = \Theta(n^2)$$

with $c_1 = \frac{1}{2}$, $c_2 = \frac{7}{2}$ and $n_1 = 1$

- Then, according to (4), we have

$$n^2 (2 - \sin n) + n^2 = \Theta(n^2),$$

with the constants $\frac{3}{2}$, $\frac{9}{2}$ and $n_1 = 1$

The converse does not hold

- By the implication (4), «$f(n) = \Theta\big(g(n)\big)$» is a sufficient condition for «$f(n) + g(n) = \Theta\big(g(n)\big)$», and «$f(n) + g(n) = \Theta\big(g(n)\big)$» is a necessary condition for «$f(n) = \Theta\big(g(n)\big)$»
- By contraposition of (4) we also have

$$f(n) + g(n) \neq \Theta\big(g(n)\big) \implies f(n) \neq \Theta\big(g(n)\big).$$

- The implication (4) does not work in the other direction (see the next counterexample), that is,

$$f(n) + g(n) = \Theta\big(g(n)\big) \not\Rightarrow f(n) = \Theta\big(g(n)\big).$$

Example: the converse of Observation 2 is false (1/4)

Let $f(n) = 10\,n$ and $g(n) = \frac{1}{10}\,n^2$.

- To prove that

$$f(n) + g(n) = 10\,n + \frac{1}{10}\,n^2 = \Theta\left(\frac{1}{10}\,n^2\right) = \Theta(g(n))$$

we determine $c_1, c_2 \in \mathbb{R}_{>0}$ and $n_1 \in \mathbb{N}$ such that

$$0 \leq c_1\,\frac{1}{10}\,n^2 \leq 10\,n + \frac{1}{10}\,n^2 \leq c_2\,\frac{1}{10}\,n^2, \qquad \forall n \geq n_1.$$

- With $n \geq 1$, dividing by $\frac{1}{10}\,n^2$ and since $c_1 > 0$, this is equivalent to

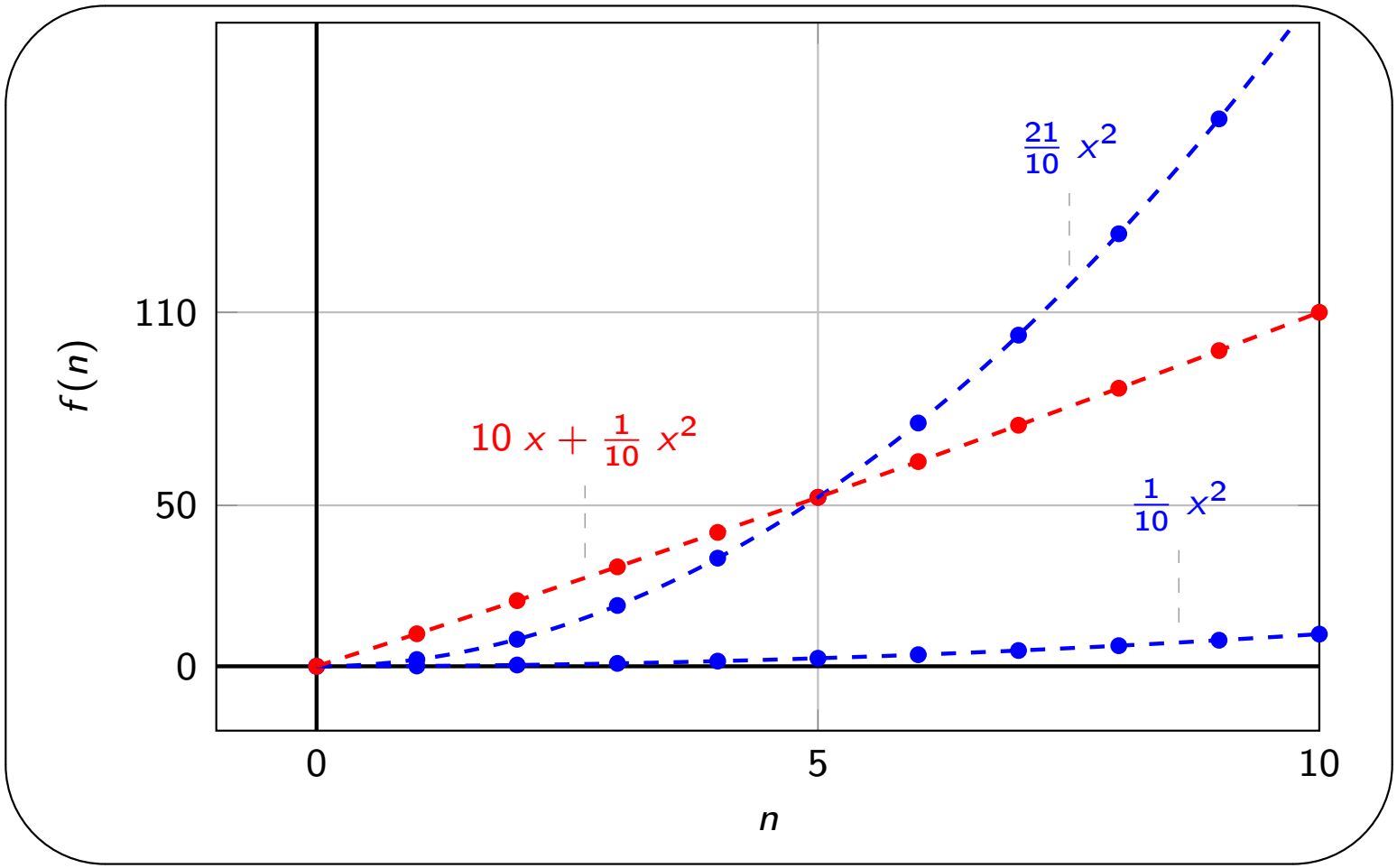
$$0 < c_1 \leq \frac{100}{n} + 1 \leq c_2, \qquad \forall n \geq n_1.$$

Example: the converse of Observation 2 is false (2/4)

- The middle term decreases with n and it is always larger than 1; for $n = 5$ it is equal to 21
- Thus, by choosing for example $c_1 = 1$, $c_2 = 21$ and $n_1 = 5$, we have

$$0 \leq \frac{1}{10}\, n^2 \leq 10\, n + \frac{1}{10}\, n^2 \leq \frac{21}{10}\, n^2, \qquad \forall n \geq 5.$$

Example: the converse of Observation 2 is false (3/4)



Example: the converse of Observation 2 is false (4/4)

- To prove that

$$f(n) = 10\,n \neq \Theta\left(\frac{1}{10}\,n^2\right) = \Theta\big(g(n)\big)$$

we proceed by contradiction and we suppose that there exist $c_1 \in \mathbb{R}_{>0}$ and $n_1 \in \mathbb{N}$ such that

$$0 \leq c_1\,\frac{1}{10}\,n^2 \leq 10\,n, \qquad \forall n \geq n_1.$$

- With $n \geq 1$, dividing by $\frac{1}{10}\,n^2$ and since $c_1 > 0$, this is equivalent to

$$0 < c_1 \leq \frac{100}{n}, \qquad \forall n \geq n_1,$$

which cannot hold for arbitrarily large n, since c_1 is a positive constant

When we add two functions,
the asymptotic behaviour of the sum
is determined by the larger of the two.

The next two observations make this precise.

Observation 3: sum and maximum (1)

Given two asymptotically non-negative functions f and g of a natural variable, we have

$$f(n) + g(n) = \Theta\big(\max\{f(n), g(n)\}\big).$$

Proof

- Since f and g are asymptotically non-negative, there exists $\tilde{n} \in \mathbb{N}$ such that $f(n) \geq 0$ and $g(n) \geq 0$ for all $n \geq \tilde{n}$
- Then the sum is at least the larger of the two terms and at most twice the larger of the two terms: for all $n \geq \tilde{n}$,

$$0 \leq \max\{f(n), g(n)\} \leq f(n) + g(n) \leq 2 \max\{f(n), g(n)\}.$$

- This shows that the constants $c_1 = 1$, $c_2 = 2$ and $n_1 = \tilde{n}$ satisfy, for all $n \geq n_1$,

$$0 \leq c_1 \max\{f(n), g(n)\} \leq f(n)+g(n) \leq c_2 \max\{f(n), g(n)\}.$$

- Therefore, $f(n) + g(n) = \Theta\big(\max\{f(n), g(n)\}\big)$

Observation 4: sum and maximum (2)

Given two asymptotically non-negative functions f and g of a natural variable, we have

$$\max\{f(n), g(n)\} = \Theta\big(f(n) + g(n)\big).$$

Proof

- Let $\tilde{n} \in \mathbb{N}$ be such that $f(n) \geq 0$ and $g(n) \geq 0$ for all $n \geq \tilde{n}$
- The larger of two non-negative numbers is at least their average and at most their sum: for all $n \geq \tilde{n}$,

$$0 \leq \frac{1}{2}\left(f(n) + g(n)\right) \leq \max\left\{f(n), g(n)\right\} \leq f(n) + g(n).$$

- This shows that the constants $c_1 = \frac{1}{2}$, $c_2 = 1$ and $n_1 = \tilde{n}$ satisfy, for all $n \geq n_1$,

$$0 \leq c_1\left(f(n) + g(n)\right) \leq \max\left\{f(n), g(n)\right\} \leq c_2\left(f(n) + g(n)\right).$$

- Therefore, $\max\{f(n), g(n)\} = \Theta\big(f(n) + g(n)\big)$

□

Example: sum and maximum

- By Observations 3 and 4 we have

$$\frac{1}{2}\,n^2 + 2^n = \Theta\left(\max\left\{\frac{1}{2}\,n^2, 2^n\right\}\right)$$

and

$$\max\left\{\frac{1}{2}\,n^2, 2^n\right\} = \Theta\left(\frac{1}{2}\,n^2 + 2^n\right).$$

- In this case $\max\left\{\frac{1}{2}\,n^2, 2^n\right\} = 2^n$ for every $n \in \mathbb{N}$, so that

$$\frac{1}{2}\,n^2 + 2^n = \Theta(2^n),$$

with $c_1 = 1$, $c_2 = 2$ and $n_1 = 0$

Summary (1/2)

Definitions:

- $f(n) = \Theta\big(g(n)\big)$: $0 \leq c_1 g(n) \leq f(n) \leq c_2 g(n)$ for all $n \geq n_1$ (asymptotic tight bound)
- $f(n) = \mathcal{O}\big(g(n)\big)$: $0 \leq f(n) \leq c\, g(n)$ (asymptotic upper bound)
- $f(n) = \Omega\big(g(n)\big)$: $0 \leq c\, g(n) \leq f(n)$ (asymptotic lower bound)

Proofs:

- To prove a statement we determine the constants and the threshold
- To disprove it we suppose that the constants exist and derive a contradiction for large n

Summary (2/2)

Relations:

- Θ is the same as $\mathcal{O}$ and Ω together; $\mathcal{O}$ or Ω alone does not give Θ ($3n^2$ and n^3, $n^2 \sin^2 n$ and n^2, $2n^3$ and n^2)

Sums:

- $f(n) = \Theta\big(g(n)\big)$ implies $f(n) + g(n) = \Theta\big(g(n)\big)$, but not conversely ($10n$ and $\frac{1}{10}n^2$)
- For asymptotically non-negative functions, $f(n) + g(n)$ and $\max\{f(n), g(n)\}$ are each Θ of the other

o and ω notations

Introduction: o and ω notations

- The asymptotic bounds expressed by the $\mathcal{O}$- and Ω-notations may or may not be asymptotically tight
- We use the o-notation to denote an asymptotic upper bound that is **not asymptotically tight**
- We use the ω-notation to denote an asymptotic lower bound that is **not asymptotically tight**
- They express that a function becomes negligible, or arbitrarily large, with respect to another one

Example: tight and non-tight upper bounds

- We have

$$3n^3 = \mathcal{O}(n^3) \qquad \text{and} \qquad 3n^3 = \Theta(n^3) :$$

n^3 is an asymptotic upper bound for $3n^3$, and it is also an asymptotically tight bound

- We have

$$3n^3 = \mathcal{O}(n^4) \qquad \text{but} \qquad 3n^3 \neq \Theta(n^4) :$$

n^4 is an asymptotic upper bound for $3n^3$, but it is not an asymptotically tight bound

Definition 5: o-notation

Given two functions f and g of a natural variable, we write $f(n) = o\big(g(n)\big)$ if for all $c \in \mathbb{R}_{>0}$ there exists $n_1 \in \mathbb{N}$ such that

$$0 \leq f(n) < c\, g(n), \qquad \forall n \geq n_1. \tag{5}$$

- If $f(n) = o\big(g(n)\big)$, then, for sufficiently large values of n, the function f is bounded from above by the function g scaled by **any** positive constant, that is, the function f becomes insignificant relative to the function g

Asymptotic upper bound that is not tight

- If $f(n) = o\big(g(n)\big)$, we say that the function g is an **asymptotic upper bound that is not tight** for the function f
- The difference with the $\mathcal{O}$-notation is in the quantifier on the constant:
  - for $\mathcal{O}$ the inequality must hold for **one** constant c
  - for o it must hold for **every** constant c
- The threshold n_1 depends on c: the smaller c, the larger, in general, n_1

Definition 6: ω-notation

Given two functions f and g of a natural variable, we write $f(n) = \omega\big(g(n)\big)$ if for all $c \in \mathbb{R}_{>0}$ there exists $n_1 \in \mathbb{N}$ such that

$$0 \leq c\, g(n) < f(n), \qquad \forall n \geq n_1. \tag{6}$$

- If $f(n) = \omega\big(g(n)\big)$, then, for sufficiently large values of n, the function f is bounded from below by the function g scaled by **any** positive constant, that is, the function f becomes arbitrarily large relative to the function g
- In this case we say that the function g is an **asymptotic lower bound that is not tight** for the function f

The two functions are asymptotically non-negative

- If $f(n) = o\big(g(n)\big)$ or $f(n) = \omega\big(g(n)\big)$ then, since $c > 0$, we have

$$g(n) \geq 0 \quad \text{and} \quad f(n) \geq 0, \qquad \forall n \geq n_1$$

- Therefore, the two functions f and g are **asymptotically non-negative**

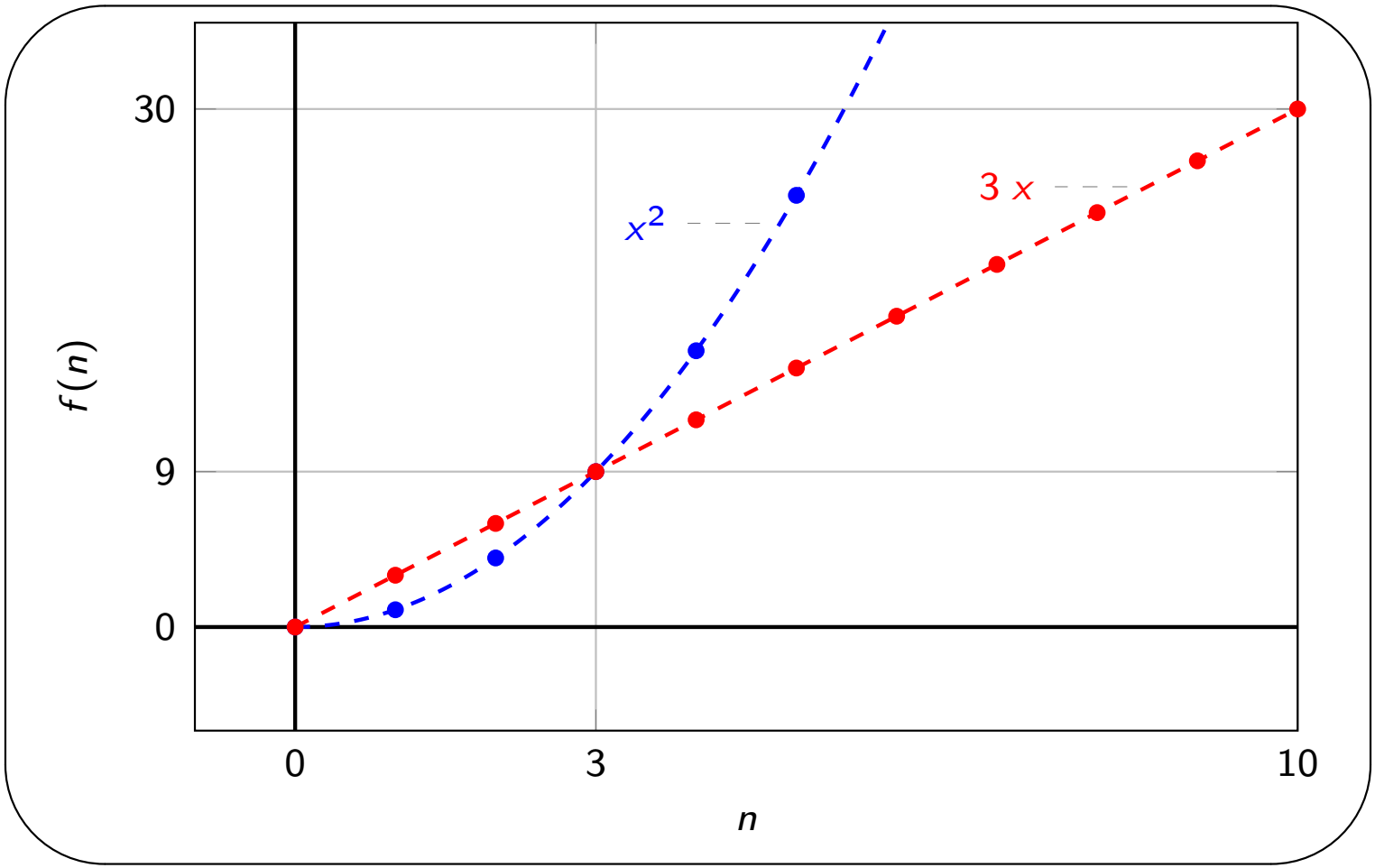
Example: upper bound that is not tight (1/4)

- To prove that
$$3n = o(n^2)$$
we need to verify that for every $c \in \mathbb{R}_{>0}$ there exists $n_1 \in \mathbb{N}$ such that
$$0 \leq 3n < c\, n^2, \qquad \forall n \geq n_1.$$
- For $n \geq 1$, dividing by n^2, the condition is equivalent to
$$0 \leq \frac{3}{n} < c, \qquad \forall n \geq n_1.$$

Example: upper bound that is not tight (2/4)

- This holds whenever $n > \frac{3}{c}$, so we can choose any $n_1 > \frac{3}{c}$, for example
$$n_1 = \left\lfloor \frac{3}{c} \right\rfloor + 1.$$
- For instance:
  - $c = 1$ gives $n_1 = 4$
  - $c = \frac{1}{2}$ gives $n_1 = 7$
  - $c = \frac{1}{10}$ gives $n_1 = 31$

Example: upper bound that is not tight (3/4)



Example: upper bound that is not tight (4/4)

- The previous proof also proves that $3n = \mathcal{O}(n^2)$ (it is enough to take one value of c, for example $c = 1$ with $n_1 = 4$)
- To prove that

$$3\,n \neq \Theta(n^2)$$

we proceed by contradiction and we suppose that there exist $c_1 \in \mathbb{R}_{>0}$ and $n_1 \in \mathbb{N}$ such that

$$0 \leq c_1\, n^2 \leq 3\, n, \qquad \forall n \geq n_1.$$

- With $n \geq 1$, dividing by n^2 and since $c_1 > 0$, this is equivalent to

$$0 < c_1 \leq \frac{3}{n}, \qquad \forall n \geq n_1,$$

which cannot hold for arbitrarily large n, since c_1 is a positive constant

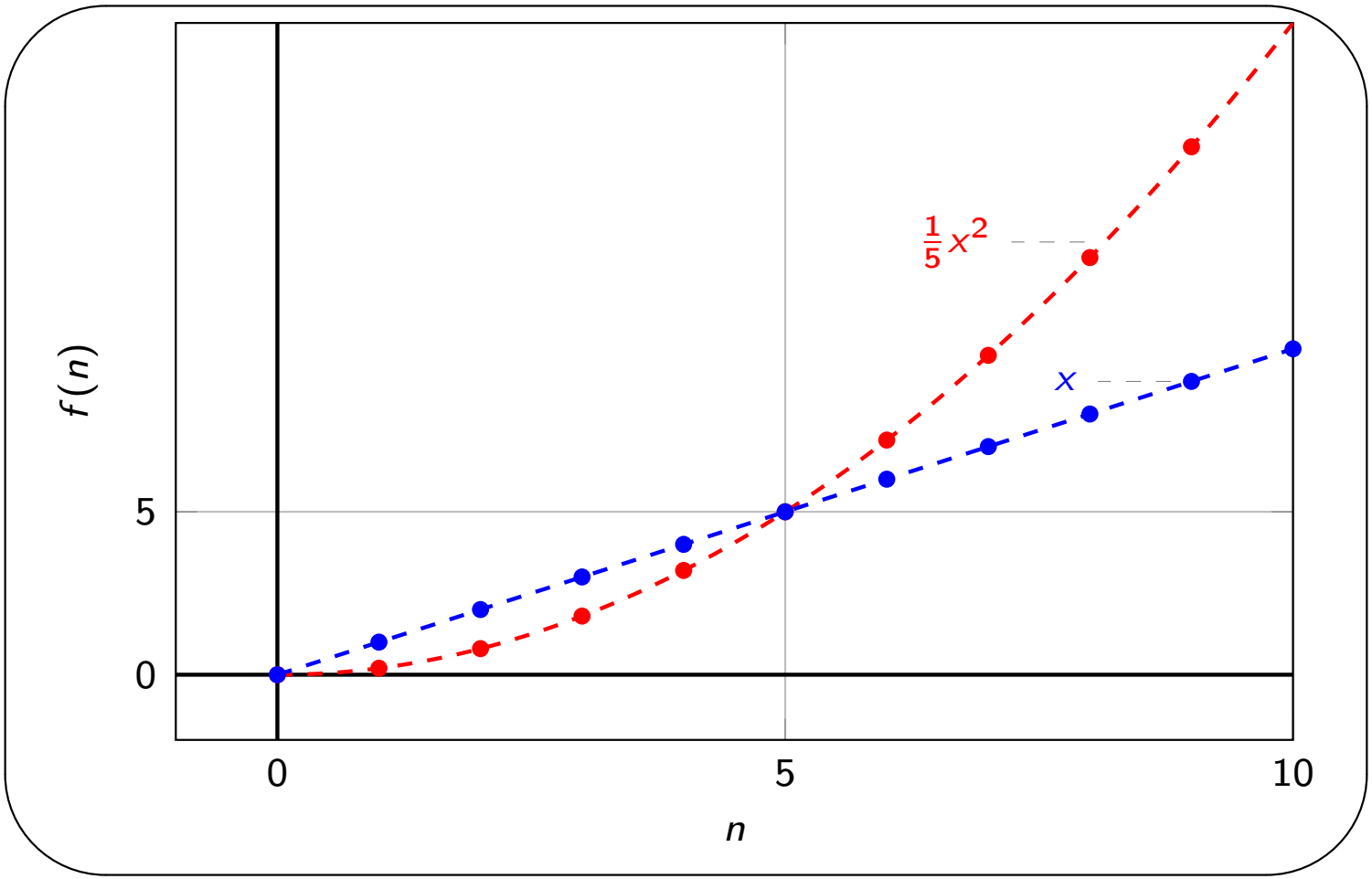
Example: lower bound that is not tight (1/4)

- To prove that
$$\frac{1}{5}\, n^2 = \omega(n)$$
we need to verify that for every $c \in \mathbb{R}_{>0}$ there exists $n_1 \in \mathbb{N}$ such that
$$0 \leq c\, n < \frac{1}{5}\, n^2, \qquad \forall n \geq n_1.$$
- For $n \geq 1$, dividing by n, the condition is equivalent to
$$0 < c < \frac{1}{5}\, n, \qquad \forall n \geq n_1.$$

Example: lower bound that is not tight (2/4)

- This holds whenever $n > 5c$, so we can choose any $n_1 > 5c$, for example
$$n_1 = \lfloor 5c \rfloor + 1.$$
- For instance:
  - $c = 1$ gives $n_1 = 6$
  - $c = 10$ gives $n_1 = 51$

Example: lower bound that is not tight (3/4)



Example: lower bound that is not tight (4/4)

- The previous proof also proves that $\frac{1}{5}\, n^2 = \Omega(n)$
- To prove that

$$\frac{1}{5}\, n^2 \neq \Theta(n)$$

we proceed by contradiction and we suppose that there exist $c_2 \in \mathbb{R}_{>0}$ and $n_1 \in \mathbb{N}$ such that

$$0 \leq \frac{1}{5}\, n^2 \leq c_2\, n, \qquad \forall n \geq n_1.$$

- With $n \geq 1$, dividing by n, this is equivalent to

$$0 \leq \frac{n}{5} \leq c_2, \qquad \forall n \geq n_1,$$

which cannot hold for arbitrarily large n, since c_2 is a constant

Summary

- $f(n) = o(g(n))$: for **every** $c > 0$ there is n_1 with $0 \leq f(n) < c\, g(n)$ for all $n \geq n_1$ (f becomes insignificant relative to g)
- $f(n) = \omega(g(n))$: for **every** $c > 0$ there is n_1 with $0 \leq c\, g(n) < f(n)$ for all $n \geq n_1$ (f becomes arbitrarily large relative to g)
- The threshold n_1 depends on c: for $3n = o(n^2)$ we can take $n_1 = \lfloor 3/c \rfloor + 1$, for $\frac{1}{5}n^2 = \omega(n)$ we can take $n_1 = \lfloor 5c \rfloor + 1$

Limits and asymptotic notations

Introduction: Limits and asymptotic notations

- When the ratio $\frac{f(n)}{g(n)}$ has a limit, its value tells us immediately which notations hold
- In this section the functions f and g are **asymptotically positive**, so that the ratio is defined for all sufficiently large n
- The limits are computed with the rules of Mathematical Analysis 1
- We arrive at the **limit criterion**, the most practical tool to compare two functions, and we see what to do when the limit does not exist

We start with the two non-tight notations:
o corresponds to the limit 0^+,
ω corresponds to the limit $+\infty$.

Observation 5: limit equal to zero

Given two asymptotically positive functions f and g of a natural variable, we have

$$\lim_{n \to +\infty} \frac{f(n)}{g(n)} = 0^+ \iff f(n) = o\big(g(n)\big). \qquad (7)$$

Proof (1/2)

Let $\tilde{n} \in \mathbb{N}$ be such that $f(n) > 0$ and $g(n) > 0$ for all $n \geq \tilde{n}$.

- If $\lim_{n \to +\infty} \dfrac{f(n)}{g(n)} = 0^+$, then for all $\varepsilon \in \mathbb{R}_{>0}$ there exists $n(\varepsilon) \in \mathbb{N}$, which we can take with $n(\varepsilon) \geq \tilde{n}$, such that

$$0 \leq \frac{f(n)}{g(n)} < \varepsilon, \quad \forall n \geq n(\varepsilon)$$

$$\implies \quad 0 \leq f(n) < \varepsilon \, g(n), \quad \forall n \geq n(\varepsilon),$$

  where we multiplied by $g(n) > 0$
- By choosing $c = \varepsilon$ and $n_1 = n(\varepsilon)$, this is the definition (5), so $f(n) = o\big(g(n)\big)$

Proof (2/2)

- If $f(n) = o\big(g(n)\big)$, then for all $c \in \mathbb{R}_{>0}$ there exists $n_1 \in \mathbb{N}$, which we can take with $n_1 \geq \tilde{n}$, such that

$$0 \leq f(n) < c\, g(n), \quad \forall n \geq n_1$$

$$\Longrightarrow \qquad 0 \leq \frac{f(n)}{g(n)} < c, \quad \forall n \geq n_1,$$

  where we divided by $g(n) > 0$

- Since c is arbitrary, this is the definition of

$$\lim_{n \to +\infty} \frac{f(n)}{g(n)} = 0^+.$$

□

Necessary and sufficient condition

- If f and g are asymptotically positive, «$\lim_{n \to +\infty} \frac{f(n)}{g(n)} = 0^+$» is a **necessary and sufficient condition** for «$f(n) = o\big(g(n)\big)$»
- We now compare o with $\mathcal{O}$: every non-tight upper bound is, in particular, an upper bound

Observation 6: o implies $\mathcal{O}$

Given two functions f and g of a natural variable, we have

$$f(n) = o\big(g(n)\big) \implies f(n) = \mathcal{O}\big(g(n)\big). \tag{8}$$

Proof.

- The definition (5) holds for every $c \in \mathbb{R}_{>0}$; in particular it holds for $c = 1$: there exists $n_1 \in \mathbb{N}$ such that

$$0 \leq f(n) < g(n) \leq 1 \cdot g(n), \qquad \forall n \geq n_1$$

- This is the definition (2) with $c = 1$

□

$\mathcal{O}$ does not imply o

- It follows that «$f(n) = o\big(g(n)\big)$» is a sufficient condition for «$f(n) = \mathcal{O}\big(g(n)\big)$», and «$f(n) = \mathcal{O}\big(g(n)\big)$» is a necessary condition for «$f(n) = o\big(g(n)\big)$»
- By contraposition of (8) we also have

$$f(n) \neq \mathcal{O}\big(g(n)\big) \implies f(n) \neq o\big(g(n)\big).$$

- The implication does not work in the other direction (see the next counterexample), that is,

$$f(n) = \mathcal{O}\big(g(n)\big) \not\Rightarrow f(n) = o\big(g(n)\big).$$

Example: an upper bound that is not an o (1/2)

- To prove that $3\,n^3 = \mathcal{O}(n^3)$ we determine $c \in \mathbb{R}_{>0}$ and $n_1 \in \mathbb{N}$ such that

$$0 \leq 3\,n^3 \leq c\,n^3, \qquad \forall n \geq n_1.$$

With $n \geq 1$, dividing by n^3, this is equivalent to

$$0 \leq 3 \leq c, \qquad \forall n \geq n_1.$$

Thus, by choosing for example $c = \frac{7}{2}$ and $n_1 = 1$, we have

$$0 \leq 3\,n^3 \leq \frac{7}{2}\,n^3, \qquad \forall n \geq 1.$$

Example: an upper bound that is not an o (2/2)

- To prove that $3\,n^3 \neq o(n^3)$ we compute the limit

$$\lim_{n \to +\infty} \frac{3\,n^3}{n^3} = 3 \neq 0^+$$

and we apply (7)

Observation 7: limit equal to $+\infty$

Given two asymptotically positive functions f and g of a natural variable, we have

$$\lim_{n\to+\infty} \frac{f(n)}{g(n)} = +\infty \iff f(n) = \omega\big(g(n)\big). \qquad (9)$$

Proof (1/2)

Let $\tilde{n} \in \mathbb{N}$ be such that $f(n) > 0$ and $g(n) > 0$ for all $n \geq \tilde{n}$.

- If $\lim_{n \to +\infty} \frac{f(n)}{g(n)} = +\infty$, then for all $\lambda \in \mathbb{R}_{>0}$ there exists $n(\lambda) \in \mathbb{N}$, which we can take with $n(\lambda) \geq \tilde{n}$, such that

$$\frac{f(n)}{g(n)} > \lambda, \quad \forall n \geq n(\lambda)$$

$$\implies \qquad 0 \leq \lambda\, g(n) < f(n), \quad \forall n \geq n(\lambda)$$

- By choosing $c = \lambda$ and $n_1 = n(\lambda)$, this is the definition (6), so $f(n) = \omega\big(g(n)\big)$

Proof (2/2)

- If $f(n) = \omega\big(g(n)\big)$, then for all $c \in \mathbb{R}_{>0}$ there exists $n_1 \in \mathbb{N}$, which we can take with $n_1 \geq \tilde{n}$, such that

$$0 \leq c\, g(n) < f(n), \quad \forall n \geq n_1$$

$$\Longrightarrow \qquad \frac{f(n)}{g(n)} > c, \quad \forall n \geq n_1$$

- Since c is arbitrary, this is the definition of

$$\lim_{n \to +\infty} \frac{f(n)}{g(n)} = +\infty.$$



Necessary and sufficient condition

- If f and g are asymptotically positive, «$\lim_{n\to+\infty} \frac{f(n)}{g(n)} = +\infty$» is a **necessary and sufficient condition** for «$f(n) = \omega\big(g(n)\big)$»
- We now compare ω with Ω: every non-tight lower bound is, in particular, a lower bound

Observation 8: ω implies Ω

Given two functions f and g of a natural variable, we have

$$f(n) = \omega\big(g(n)\big) \quad \Longrightarrow \quad f(n) = \Omega\big(g(n)\big). \qquad (10)$$

Proof.

- The definition (6) holds for every $c \in \mathbb{R}_{>0}$; in particular it holds for $c = 1$: there exists $n_1 \in \mathbb{N}$ such that

$$0 \leq 1 \cdot g(n) < f(n), \qquad \forall n \geq n_1$$

- This is the definition (3) with $c = 1$



Ω does not imply ω

- It follows that «$f(n) = \omega\big(g(n)\big)$» is a sufficient condition for «$f(n) = \Omega\big(g(n)\big)$», and «$f(n) = \Omega\big(g(n)\big)$» is a necessary condition for «$f(n) = \omega\big(g(n)\big)$»
- By contraposition of (10) we also have

$$f(n) \neq \Omega\big(g(n)\big) \implies f(n) \neq \omega\big(g(n)\big).$$

- The implication does not work in the other direction (see the next counterexample), that is,

$$f(n) = \Omega\big(g(n)\big) \not\Rightarrow f(n) = \omega\big(g(n)\big).$$

Example: a lower bound that is not an ω (1/2)

- To prove that $3\,n^3 = \Omega(n^3)$ we determine $c \in \mathbb{R}_{>0}$ and $n_1 \in \mathbb{N}$ such that

$$0 \leq c\,n^3 \leq 3\,n^3, \qquad \forall n \geq n_1.$$

With $n \geq 1$, dividing by n^3 and since $c > 0$, this is equivalent to

$$0 < c \leq 3, \qquad \forall n \geq n_1.$$

Thus, by choosing for example $c = \frac{5}{2}$ and $n_1 = 1$, we have

$$0 \leq \frac{5}{2}\,n^3 \leq 3\,n^3, \qquad \forall n \geq 1.$$

Example: a lower bound that is not an ω (2/2)

- To prove that $3\,n^3 \neq \omega(n^3)$ we compute the limit

$$\lim_{n \to +\infty} \frac{3\,n^3}{n^3} = 3 \neq +\infty$$

  and we apply (9)

We now turn to Θ, $\mathcal{O}$ and Ω:
a finite positive limit gives Θ.

Observation 9: finite positive limit

Given two asymptotically positive functions f and g of a natural variable, we have

$$\lim_{n\to+\infty} \frac{f(n)}{g(n)} = \ell \in \mathbb{R}_{>0} \quad \Longrightarrow \quad f(n) = \Theta\big(g(n)\big). \qquad (11)$$

Proof

- If $\lim_{n \to +\infty} \frac{f(n)}{g(n)} = \ell \in \mathbb{R}_{>0}$, then for all $\varepsilon \in \mathbb{R}_{>0}$ there exists $n(\varepsilon) \in \mathbb{N}$ such that $g(n) > 0$ and

$$\ell - \varepsilon < \frac{f(n)}{g(n)} < \ell + \varepsilon, \qquad \forall n \geq n(\varepsilon).$$

- Since $\ell > 0$, we can choose ε small enough to have $\ell - \varepsilon > 0$, for example $\varepsilon = \frac{\ell}{2}$
- By setting $c_1 = \ell - \varepsilon = \frac{\ell}{2} > 0$ and $c_2 = \ell + \varepsilon = \frac{3\ell}{2} > 0$, and multiplying by $g(n) > 0$, we have

$$0 < c_1\, g(n) < f(n) < c_2\, g(n), \qquad \forall n \geq n(\varepsilon),$$

and, as a consequence, $f(n) = \Theta\big(g(n)\big)$ with $n_1 = n(\varepsilon)$



Θ without a limit (1/2)

- It follows that «$\lim_{n\to+\infty} \frac{f(n)}{g(n)} = \ell \in \mathbb{R}_{>0}$» is a sufficient condition for «$f(n) = \Theta\big(g(n)\big)$», and «$f(n) = \Theta\big(g(n)\big)$» is a necessary condition for «$\lim_{n\to+\infty} \frac{f(n)}{g(n)} = \ell \in \mathbb{R}_{>0}$»
- By contraposition of (11) we also have

$$f(n) \neq \Theta\big(g(n)\big) \implies \text{the limit of } \frac{f(n)}{g(n)}$$

$$\text{is not a number } \ell \in \mathbb{R}_{>0}.$$

Θ without a limit (2/2)

- The implication does not work in the other direction. A counterexample is given by an earlier example:

$$n^2\,(2-\sin n)=\Theta(n^2),$$

but

$$\lim_{n\to+\infty}\frac{n^2\,(2-\sin n)}{n^2}=\lim_{n\to+\infty}(2-\sin n)\text{ does not exist.}$$

- Accordingly, we have

$$f(n)=\Theta\big(g(n)\big)\quad\not\Rightarrow\quad\lim_{n\to+\infty}\frac{f(n)}{g(n)}=\ell\in\mathbb{R}_{>0}.$$

When the ratio has no limit

It is possible that $f(n) = \Theta\big(g(n)\big)$
even though $\frac{f(n)}{g(n)}$ does not converge to a limit.

In this case the limit criterion cannot be applied,
and we use the definition.

Example: asymptotic tight bound without a limit (1/4)

- To prove that
$$(2 + (-1)^n)\, n^2 = \Theta(n^2)$$
we determine $c_1, c_2 \in \mathbb{R}_{>0}$ and $n_1 \in \mathbb{N}$ such that
$$0 \leq c_1\, n^2 \leq (2 + (-1)^n)\, n^2 \leq c_2\, n^2, \qquad \forall n \geq n_1.$$
- We have
$$2 + (-1)^n = \begin{cases} 1 & \text{if } n \text{ is odd,} \\ 3 & \text{if } n \text{ is even.} \end{cases}$$

Example: asymptotic tight bound without a limit (2/4)

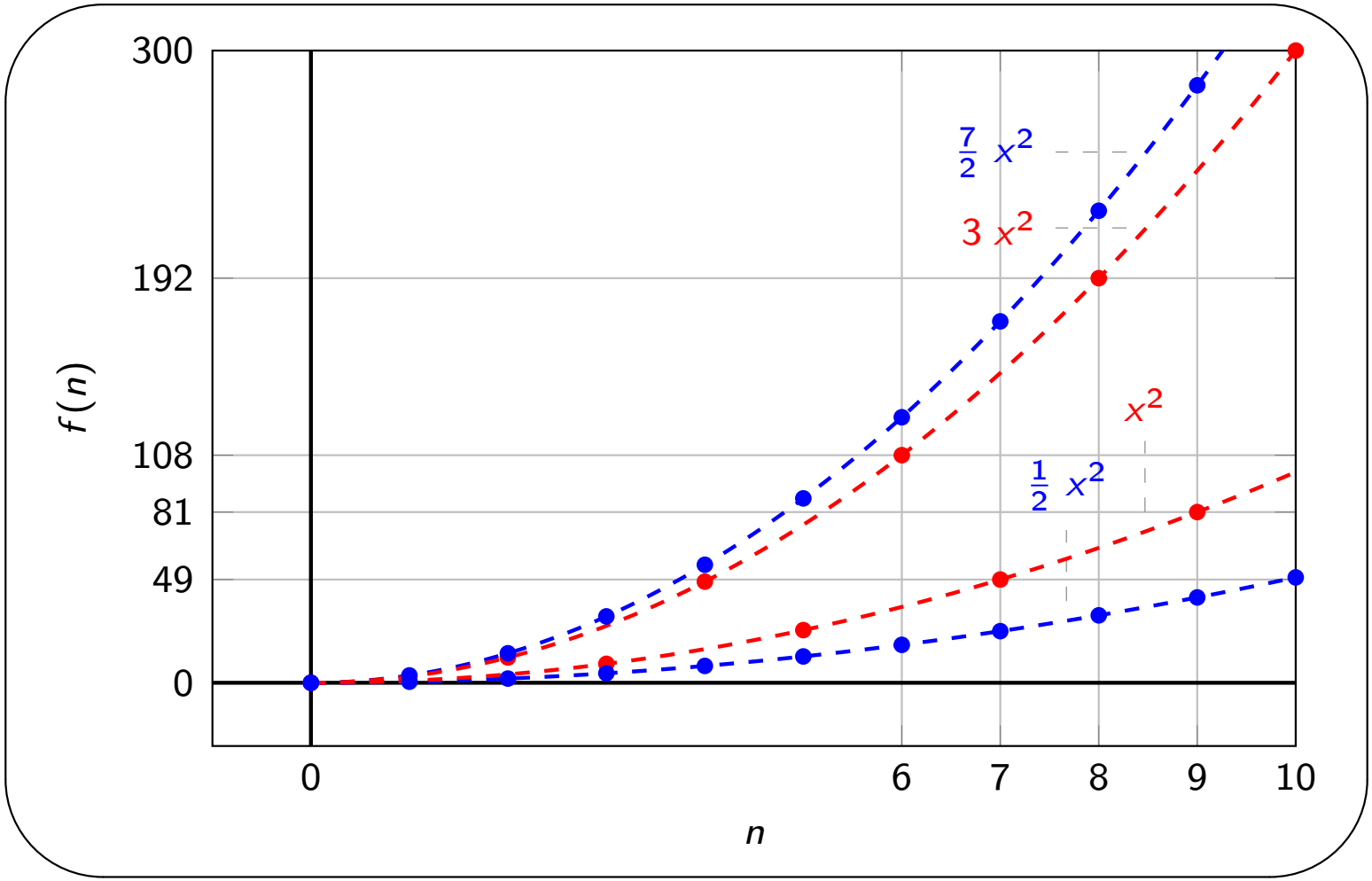
- Hence

$$(2 + (-1)^n)\, n^2 = \begin{cases} n^2 & \text{if } n \text{ is odd,} \\ 3\, n^2 & \text{if } n \text{ is even.} \end{cases}$$

- Thus, by choosing for example $c_1 = \frac{1}{2}$, $c_2 = \frac{7}{2}$ and $n_1 = 1$, we have

$$0 \leq \frac{1}{2}\, n^2 \leq (2 + (-1)^n)\, n^2 \leq \frac{7}{2}\, n^2, \qquad \forall n \geq 1.$$

Example: asymptotic tight bound without a limit (3/4)



Example: asymptotic tight bound without a limit (4/4)

- On the other hand,

$$\lim_{n\to+\infty} \frac{(2+(-1)^n)\, n^2}{n^2} = \lim_{n\to+\infty} (2+(-1)^n) \text{ does not exist,}$$

since the ratio is 1 for all odd n and 3 for all even n

- The bound $\Theta(n^2)$ holds, but it can only be proved with the definition

Observation 10: limit criterion for $\mathcal{O}$

Given two asymptotically positive functions f and g of a natural variable, we have

$$\lim_{n\to+\infty} \frac{f(n)}{g(n)} = \ell \in \mathbb{R}_{>0} \quad \text{or} \quad 0^+ \implies \quad f(n) = \mathcal{O}\big(g(n)\big). \tag{12}$$

Proof

By Observations 9, 1, 5 and 6, we have

$$\lim_{n\to+\infty} \frac{f(n)}{g(n)} = \ell \in \mathbb{R}_{>0} \implies f(n) = \Theta\big(g(n)\big)$$
$$\implies f(n) = \mathcal{O}\big(g(n)\big),$$

$$\lim_{n\to+\infty} \frac{f(n)}{g(n)} = 0^+ \iff f(n) = o\big(g(n)\big)$$
$$\implies f(n) = \mathcal{O}\big(g(n)\big).$$

In both cases $f(n) = \mathcal{O}\big(g(n)\big)$. □

$\mathcal{O}$ without a limit (1/2)

- It follows that «$\lim_{n\to+\infty} \frac{f(n)}{g(n)} = \ell \in \mathbb{R}_{>0}$ or 0^+» is a sufficient condition for «$f(n) = \mathcal{O}\big(g(n)\big)$»
- By contraposition of (12) we also have

$$f(n) \neq \mathcal{O}\big(g(n)\big) \implies \text{the limit of } \frac{f(n)}{g(n)} \text{ is neither}$$

$$\text{a number } \ell \in \mathbb{R}_{>0} \text{ nor } 0^+.$$

$\mathcal{O}$ without a limit (2/2)

- The implication does not work in the other direction. A counterexample is given by an earlier example:

$$n^2 \ \sin^2 n = \mathcal{O}(n^2),$$

but

$$\lim_{n\to+\infty} \frac{n^2 \ \sin^2 n}{n^2} = \lim_{n\to+\infty} \sin^2 n \text{ does not exist.}$$

- Accordingly, we have

$$f(n) = \mathcal{O}\big(g(n)\big) \quad \not\Rightarrow \quad \lim_{n\to+\infty} \frac{f(n)}{g(n)} = \ell \in \mathbb{R}_{>0} \text{ or } 0^+.$$

When the ratio has no limit

It is possible that $f(n) = \mathcal{O}\big(g(n)\big)$
even though $\frac{f(n)}{g(n)}$ does not converge to a limit.

Observation 11: limit criterion for Ω

Given two asymptotically positive functions f and g of a natural variable, we have

$$\lim_{n \to +\infty} \frac{f(n)}{g(n)} = \ell \in \mathbb{R}_{>0} \quad or \quad +\infty \implies f(n) = \Omega\big(g(n)\big). \tag{13}$$

Proof

By Observations 9, 1, 7 and 8, we have

$$\lim_{n\to+\infty} \frac{f(n)}{g(n)} = \ell \in \mathbb{R}_{>0} \implies f(n) = \Theta\big(g(n)\big)$$
$$\implies f(n) = \Omega\big(g(n)\big),$$

$$\lim_{n\to+\infty} \frac{f(n)}{g(n)} = +\infty \iff f(n) = \omega\big(g(n)\big)$$
$$\implies f(n) = \Omega\big(g(n)\big).$$

In both cases $f(n) = \Omega\big(g(n)\big)$. □

Ω without a limit (1/2)

- It follows that «$\lim_{n\to+\infty} \frac{f(n)}{g(n)} = \ell \in \mathbb{R}_{>0}$ or $+\infty$» is a sufficient condition for «$f(n) = \Omega\big(g(n)\big)$»
- By contraposition of (13) we also have

$$f(n) \neq \Omega\big(g(n)\big) \implies \text{the limit of } \frac{f(n)}{g(n)} \text{ is neither}$$

$$\text{a number } \ell \in \mathbb{R}_{>0} \text{ nor } +\infty.$$

Ω without a limit (2/2)

- The implication does not work in the other direction. A counterexample is given by the previous example: since $(2 + (-1)^n)\, n^2 = \Theta(n^2)$, by Observation 1 we have

$$(2 + (-1)^n)\, n^2 = \Omega(n^2),$$

but

$$\lim_{n \to +\infty} \frac{(2 + (-1)^n)\, n^2}{n^2} = \lim_{n \to +\infty} (2 + (-1)^n) \text{ does not exist.}$$

- Accordingly, we have

$$f(n) = \Omega\big(g(n)\big) \quad \not\Rightarrow \quad \lim_{n \to +\infty} \frac{f(n)}{g(n)} = \ell \in \mathbb{R}_{>0} \text{ or } +\infty.$$

When the ratio has no limit

It is possible that $f(n) = \Omega\big(g(n)\big)$
even though $\frac{f(n)}{g(n)}$ does not converge to a limit.

Example: asymptotic bounds from limits (1/3)

- We have

$$\lim_{n\to+\infty} \frac{6\,n - n^3 + 2\,n^4}{n^4} = \lim_{n\to+\infty} \left(\underbrace{\frac{6}{n^3}}_{\to 0} - \underbrace{\frac{1}{n}}_{\to 0} + 2 \right)$$
$$= 2 \in \mathbb{R}_{>0}.$$

- Then:
  - according to (12), we have $6\,n - n^3 + 2\,n^4 = \mathcal{O}(n^4)$
  - according to (13), we have $6\,n - n^3 + 2\,n^4 = \Omega(n^4)$
  - finally, according to (11), we also have $6\,n - n^3 + 2\,n^4 = \Theta(n^4)$

Example: asymptotic bounds from limits (2/3)

- We have

$$\lim_{n\to+\infty} \frac{n + 3\ n^3}{n^4} = \lim_{n\to+\infty} \left(\underbrace{\frac{1}{n^3}}_{\to 0} + \underbrace{\frac{3}{n}}_{\to 0} \right) = 0^+.$$

- Then:
  - according to (7), we have $n + 3\ n^3 = o(n^4)$
  - moreover, according to (8) or to (12), we have $n + 3\ n^3 = \mathcal{O}(n^4)$

Example: asymptotic bounds from limits (3/3)

- We have

$$\lim_{n \to +\infty} \frac{n + 3\ n^4}{n^3} = \lim_{n \to +\infty} \left(\underbrace{\frac{1}{n^2}}_{\to 0} + 3\ n \right) = +\infty.$$

- Then:
  - according to (9), we have $n + 3\ n^4 = \omega(n^3)$
  - moreover, according to (10) or to (13), we have $n + 3\ n^4 = \Omega(n^3)$

The limit criterion

When the limit of the ratio exists,
the previous observations can be reversed:

the value of the limit tells us exactly
which of the five notations hold
and which ones do not.

Proposition 1: limit criterion

Let f and g be two asymptotically positive functions of a natural variable such that the limit

$$L = \lim_{n \to +\infty} \frac{f(n)}{g(n)}$$

exists (equal to 0^+, to a number $\ell \in \mathbb{R}_{>0}$, or to $+\infty$). Then:

$$\begin{aligned}
f(n) = o\big(g(n)\big) &\iff L = 0^+, \\
f(n) = \mathcal{O}\big(g(n)\big) &\iff L = 0^+ \text{ or } L \in \mathbb{R}_{>0}, \\
f(n) = \Theta\big(g(n)\big) &\iff L \in \mathbb{R}_{>0}, \\
f(n) = \Omega\big(g(n)\big) &\iff L \in \mathbb{R}_{>0} \text{ or } L = +\infty, \\
f(n) = \omega\big(g(n)\big) &\iff L = +\infty.
\end{aligned}$$

Proof (1/3)

- The implications from right to left are (7), (12), (11), (13) and (9)
- The first and the last equivalences are (7) and (9)
- For the other implications we first observe that $f(n) = \mathcal{O}(g(n))$ and $f(n) = \omega(g(n))$ **cannot hold together**

Proof (2/3)

- Indeed, if $0 \leq f(n) \leq c\, g(n)$ for all $n \geq n_1'$, the definition (6) with the same constant c gives n_1'' such that $c\, g(n) < f(n)$ for all $n \geq n_1''$
- For $n \geq \max\{n_1', n_1''\}$ we would have

$$c\, g(n) < f(n) \leq c\, g(n),$$

  which is impossible
- In the same way, $f(n) = \Omega\big(g(n)\big)$ and $f(n) = o\big(g(n)\big)$ cannot hold together, since they would give

$$c\, g(n) \leq f(n) < c\, g(n)$$

- Since the limit exists, exactly one of the three cases $L = 0^+$, $L \in \mathbb{R}_{>0}$ and $L = +\infty$ holds

Proof (3/3)

- If $f(n) = \mathcal{O}\big(g(n)\big)$, then $L \neq +\infty$, since otherwise $f(n) = \omega\big(g(n)\big)$ by (9). Hence $L = 0^+$ or $L \in \mathbb{R}_{>0}$
- If $f(n) = \Omega\big(g(n)\big)$, then $L \neq 0^+$, since otherwise $f(n) = o\big(g(n)\big)$ by (7). Hence $L \in \mathbb{R}_{>0}$ or $L = +\infty$
- If $f(n) = \Theta\big(g(n)\big)$, then $f(n) = \mathcal{O}\big(g(n)\big)$ and $f(n) = \Omega\big(g(n)\big)$ by Observation 1, and by the two previous points $L \in \mathbb{R}_{>0}$

□

The limit criterion in practice (1/2)

For two asymptotically positive functions f and g whose ratio has a limit L, the limit criterion reads:

- if $L = 0^+$, then $f(n) = o\big(g(n)\big)$ and $f(n) = \mathcal{O}\big(g(n)\big)$, and the other three notations do not hold
- if $L \in \mathbb{R}_{>0}$, then $f(n) = \Theta\big(g(n)\big)$, $f(n) = \mathcal{O}\big(g(n)\big)$ and $f(n) = \Omega\big(g(n)\big)$, while $f(n) \neq o\big(g(n)\big)$ and $f(n) \neq \omega\big(g(n)\big)$
- if $L = +\infty$, then $f(n) = \omega\big(g(n)\big)$ and $f(n) = \Omega\big(g(n)\big)$, and the other three notations do not hold

The limit criterion in practice (2/2)

If the ratio has no limit,
the criterion cannot be applied,
and we use the definitions.

(As in the examples $n^2\,(2-\sin n) = \Theta(n^2)$ and
$(2+(-1)^n)\,n^2 = \Theta(n^2)$.)

Example: the limit criterion (1/2)

- In the course *Mathematical Analysis 1* (chapter *Hierarchies of infinities and the ratio test*) it is proved that

$$\lim_{n \to +\infty} \frac{\ln n}{n} = 0^+.$$

- By Proposition 1, we have

$$\ln n = o(n) \qquad \text{and} \qquad \ln n = \mathcal{O}(n),$$

while

$$\ln n \neq \Theta(n), \qquad \ln n \neq \Omega(n) \qquad \text{and} \qquad \ln n \neq \omega(n).$$

Example: the limit criterion (2/2)

- In the same chapter of that course it is proved that $\lim_{n\to+\infty} \frac{n^2}{2^n} = 0^+$, hence

$$\lim_{n\to+\infty} \frac{2^n + n^2}{2^n} = \lim_{n\to+\infty} \left(1 + \underbrace{\frac{n^2}{2^n}}_{\to 0^+}\right) = 1 \in \mathbb{R}_{>0}.$$

- By Proposition 1, we have

$$2^n + n^2 = \Theta(2^n), \qquad 2^n + n^2 = \mathcal{O}(2^n), \qquad 2^n + n^2 = \Omega(2^n),$$

while

$$2^n + n^2 \neq o(2^n) \qquad \text{and} \qquad 2^n + n^2 \neq \omega(2^n).$$

Summary (1/2)

Limits and the non-tight notations:

- $\lim \frac{f(n)}{g(n)} = 0^+ \iff f(n) = o\big(g(n)\big)$, and o implies $\mathcal{O}$
- $\lim \frac{f(n)}{g(n)} = +\infty \iff f(n) = \omega\big(g(n)\big)$, and ω implies Ω
- Neither implication can be reversed ($3n^3$ and n^3)

Limits and Θ, $\mathcal{O}$, Ω:

- A finite positive limit gives Θ; the limit ℓ or 0^+ gives $\mathcal{O}$; the limit ℓ or $+\infty$ gives Ω

Summary (2/2)

The limit criterion:

- When the limit L of the ratio exists, it decides exactly which of the five notations hold
- $L = 0^+$: o and $\mathcal{O}$; $L \in \mathbb{R}_{>0}$: Θ, $\mathcal{O}$ and Ω; $L = +\infty$: ω and Ω

No limit:

- A bound may hold even if the ratio has no limit ($2 - \sin n$, $\sin^2 n$, $2 + (-1)^n$): then we use the definitions

Relational properties of asymptotic notations

Introduction: Relational properties of asymptotic notations

- We prove that the notations are transitive, that Θ, $\mathcal{O}$ and Ω are reflexive, that Θ is symmetric, and that $\mathcal{O}$, Ω and o, ω are transpose symmetric
- These properties make the asymptotic comparison of two functions similar to the comparison of two real numbers
- We also see an important difference: two functions may not be comparable at all

Proposition 2: relational properties

Given three functions f, g and h of a natural variable, the following properties hold.

1. ***Transitivity***:

$$f(n) = o\big(g(n)\big) \text{ and } g(n) = o\big(h(n)\big) \;\Rightarrow\; f(n) = o\big(h(n)\big),$$
$$f(n) = \mathcal{O}\big(g(n)\big) \text{ and } g(n) = \mathcal{O}\big(h(n)\big) \;\Rightarrow\; f(n) = \mathcal{O}\big(h(n)\big),$$
$$f(n) = \Theta\big(g(n)\big) \text{ and } g(n) = \Theta\big(h(n)\big) \;\Rightarrow\; f(n) = \Theta\big(h(n)\big),$$
$$f(n) = \Omega\big(g(n)\big) \text{ and } g(n) = \Omega\big(h(n)\big) \;\Rightarrow\; f(n) = \Omega\big(h(n)\big),$$
$$f(n) = \omega\big(g(n)\big) \text{ and } g(n) = \omega\big(h(n)\big) \;\Rightarrow\; f(n) = \omega\big(h(n)\big).$$

Proposition: relational properties (continued)

2. ***Reflexivity*** *(for an asymptotically non-negative function f):*

$$f(n) = \Theta\big(f(n)\big), \qquad f(n) = \mathcal{O}\big(f(n)\big), \qquad f(n) = \Omega\big(f(n)\big).$$

3. ***Symmetry****:*

$$f(n) = \Theta\big(g(n)\big) \iff g(n) = \Theta\big(f(n)\big).$$

4. ***Transpose symmetry****:*

$$f(n) = \mathcal{O}\big(g(n)\big) \iff g(n) = \Omega\big(f(n)\big),$$
$$f(n) = o\big(g(n)\big) \iff g(n) = \omega\big(f(n)\big).$$

Proof (1/6)

Transitivity.

- Let $0 \leq f(n) \leq c'\, g(n)$ for $n \geq n_1'$ and $0 \leq g(n) \leq c''\, h(n)$ for $n \geq n_1''$
- Then

$$0 \leq f(n) \leq c'c''\, h(n), \qquad \forall n \geq \max\{n_1', n_1''\},$$

  and $f(n) = \mathcal{O}\big(h(n)\big)$ with $c = c'c'' > 0$
- The proofs for Ω and Θ are the same, with the products of the constants

Proof (2/6)

Transitivity (continued).

- For o, let $c \in \mathbb{R}_{>0}$: the definition of $f(n) = o\big(g(n)\big)$ with the constant 1 and the definition of $g(n) = o\big(h(n)\big)$ with the constant c give two thresholds n_1' and n_1'' such that

$$0 \leq f(n) < g(n) < c\, h(n), \qquad \forall n \geq \max\{n_1', n_1''\}.$$

- Since c is arbitrary, $f(n) = o\big(h(n)\big)$
- The proof for ω is the same, with $0 \leq c\, h(n) < g(n) < f(n)$

Proof (3/6)

Reflexivity.

- Let $\tilde{n}$ be such that $f(n) \geq 0$ for all $n \geq \tilde{n}$. Then

$$0 \leq 1 \cdot f(n) \leq f(n) \leq 1 \cdot f(n), \qquad \forall n \geq \tilde{n}:$$

the definitions of Θ, $\mathcal{O}$ and Ω hold with all the constants equal to 1 and $n_1 = \tilde{n}$

Proof (4/6)

Symmetry.

- If $0 \leq c_1\, g(n) \leq f(n) \leq c_2\, g(n)$ for all $n \geq n_1$, dividing the left inequality by c_1 and the right inequality by c_2 we obtain

$$0 \leq \frac{1}{c_2} f(n) \leq g(n) \leq \frac{1}{c_1} f(n), \qquad \forall n \geq n_1,$$

  that is, $g(n) = \Theta\big(f(n)\big)$
- Exchanging the roles of f and g we obtain the other direction

Proof (5/6)

Transpose symmetry.

- We have $0 \leq f(n) \leq c\, g(n)$ if and only if $0 \leq \frac{1}{c} f(n) \leq g(n)$: the first inequality defines $f(n) = \mathcal{O}\big(g(n)\big)$ with the constant c, the second one defines $g(n) = \Omega\big(f(n)\big)$ with the constant $\frac{1}{c}$, and the threshold is the same

Proof (6/6)

Transpose symmetry (continued).

- For o and ω, let $f(n) = o\big(g(n)\big)$ and let $c \in \mathbb{R}_{>0}$. The definition of $f(n) = o\big(g(n)\big)$ with the constant $\frac{1}{c}$ gives n_1 such that

$$0 \leq f(n) < \frac{1}{c}\, g(n), \quad \text{that is,} \quad 0 \leq c\, f(n) < g(n),$$

  for all $n \geq n_1$
- Since c is arbitrary, $g(n) = \omega\big(f(n)\big)$. The other direction is proved in the same way

□

Asymptotic comparison and real numbers

Because these relational properties hold, we can draw an analogy between the **asymptotic comparison** of two functions and the comparison of two real numbers a and b:

$$f(n) = o\big(g(n)\big) \quad \text{is like} \quad a < b,$$

$$f(n) = \mathcal{O}\big(g(n)\big) \quad \text{is like} \quad a \leq b,$$

$$f(n) = \Theta\big(g(n)\big) \quad \text{is like} \quad a = b,$$

$$f(n) = \Omega\big(g(n)\big) \quad \text{is like} \quad a \geq b,$$

$$f(n) = \omega\big(g(n)\big) \quad \text{is like} \quad a > b.$$

An important difference: trichotomy

- For any two real numbers a and b, exactly one of the following three cases holds:

$$a < b, \qquad a = b, \qquad a > b \qquad \text{(trichotomy property)}.$$

- The analogous property **does not hold** for the asymptotic comparison of two functions

- There exist two functions f and g such that **none** of the five notations holds, as the next example shows

Example: two functions that cannot be compared (1/6)

- We consider the asymptotically positive function

$$f : \mathbb{N} \to \mathbb{R}, \qquad f : n \mapsto n^{1+\sin n}$$

and the function $g(n) = n$. For every $n \geq 1$ we have $\frac{f(n)}{g(n)} = n^{\sin n}$

- Since π is irrational, the values $n \bmod 2\pi$, $n \in \mathbb{N}$, are dense in $[0, 2\pi]$

Example: two functions that cannot be compared (2/6)

- Hence there exist arbitrarily large integers n with $\sin n > \frac{1}{2}$, for which

$$n^{\sin n} \geq n^{1/2} = \sqrt{n},$$

and arbitrarily large integers m with $\sin m < -\frac{1}{2}$, for which

$$m^{\sin m} \leq m^{-1/2} = \frac{1}{\sqrt{m}}.$$

Example: two functions that cannot be compared (3/6)

- To prove that
$$n^{1+\sin n} \neq \mathcal{O}(n),$$
we proceed by contradiction and we suppose that there exist $c \in \mathbb{R}_{>0}$ and $n_1 \in \mathbb{N}$ such that $0 \leq n^{1+\sin n} \leq c\, n$, that is, $n^{\sin n} \leq c$, for all $n \geq n_1$. This is false for the arbitrarily large integers n with $\sin n > \frac{1}{2}$ and $\sqrt{n} > c$

Example: two functions that cannot be compared (4/6)

- To prove that
$$n^{1+\sin n} \neq \Omega(n),$$
we proceed by contradiction and we suppose that there exist $c \in \mathbb{R}_{>0}$ and $n_1 \in \mathbb{N}$ such that $0 \leq c\, n \leq n^{1+\sin n}$, that is, $c \leq n^{\sin n}$, for all $n \geq n_1$. This is false for the arbitrarily large integers m with $\sin m < -\frac{1}{2}$ and $\frac{1}{\sqrt{m}} < c$

Example: two functions that cannot be compared (5/6)

- By Observation 1, it follows that

$$n^{1+\sin n} \neq \Theta(n).$$

- By the contrapositions of (8) and (10), it also follows that

$$n^{1+\sin n} \neq o(n) \qquad \text{and} \qquad n^{1+\sin n} \neq \omega(n).$$

Example: two functions that cannot be compared (6/6)

- The same conclusion follows from the limit

$$\lim_{n\to+\infty} \frac{n^{1+\sin n}}{n} = \lim_{n\to+\infty} n^{\sin n} = \lim_{n\to+\infty} e^{\sin n \ln n},$$

  which does not exist: along the integers n with $\sin n > \frac{1}{2}$ the ratio tends to $+\infty$, along the integers m with $\sin m < -\frac{1}{2}$ it tends to 0^+
- The limit is neither 0^+ nor $+\infty$, and we apply (7) and (9)
- Therefore **none of the five notations** holds between $n^{1+\sin n}$ and n
- This is a stronger failure than the failure of trichotomy: not even the analogues of $a \leq b$ or $a \geq b$ hold

Example: the graph of $n^{1+\sin n}$ (1/2)

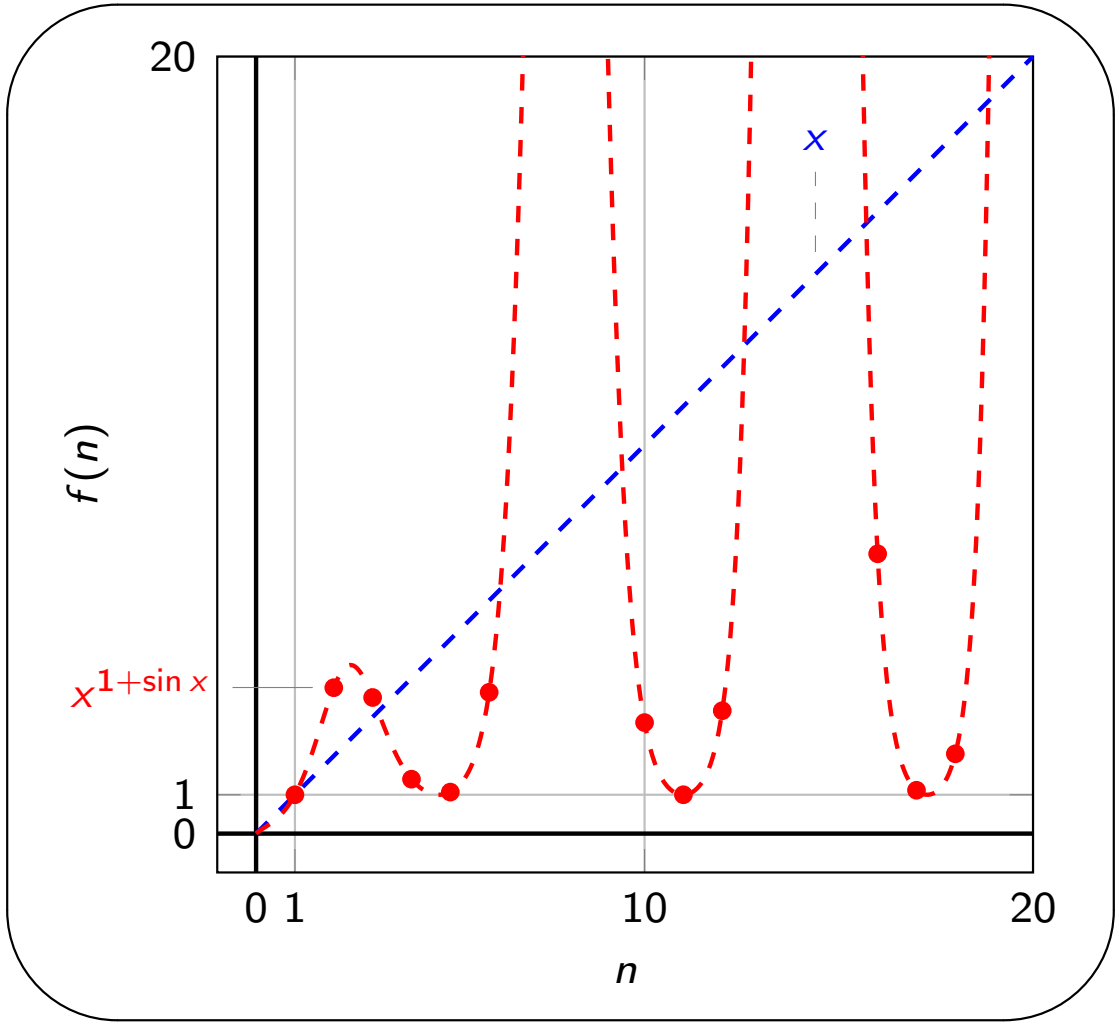
- This is the graph of the function

$$f : \mathbb{N} \to \mathbb{R}, \qquad f : n \mapsto n^{1+\sin n},$$

  together with the graph of the function $g : \mathbb{R}_{\geq 0} \to \mathbb{R}$, $g : x \mapsto x$

- The values $f(n)$ are alternately above and below the line g, farther and farther from it

Example: the graph of $n^{1+\sin n}$ (2/2)



Summary

- **Transitivity**: all five notations are transitive
- **Reflexivity**: Θ, $\mathcal{O}$ and Ω are reflexive
- **Symmetry**: Θ is symmetric
- **Transpose symmetry**:
  $f(n) = \mathcal{O}\big(g(n)\big) \iff g(n) = \Omega\big(f(n)\big)$ and
  $f(n) = o\big(g(n)\big) \iff g(n) = \omega\big(f(n)\big)$

- o, $\mathcal{O}$, Θ, Ω, ω are like $<$, $\leq$, $=$, $\geq$, $>$ between real numbers
- But trichotomy fails: $n^{1+\sin n}$ and n cannot be compared with any of the five notations

Asymptotic notation in the analysis of algorithms

Introduction: Asymptotic notation in the analysis of algorithms

- In the chapters «Input size» and «Running time» we measure the size $|I|$ of the input of an algorithm and we count the number of steps $S(n)$ that the algorithm executes in the worst case, as a function of a natural parameter n
- The exact expression of $S(n)$ depends on the conventions used to count the steps, and the time of one step depends on the computer
- The asymptotic notation keeps only what does not depend on these details: the **order of growth**

What does not matter

- **Constant factors do not matter.** If $f(n) = \Theta\big(g(n)\big)$ and $\lambda \in \mathbb{R}_{>0}$, then
$$\lambda f(n) = \Theta\big(g(n)\big),$$
with the constants λc_1 and λc_2 and the same threshold
- For example, an algorithm executing $4n$ steps and an algorithm executing n steps both have a number of steps $\Theta(n)$

- **Lower-order terms do not matter.** This is made precise by the next observation, which follows from Observation 3

Observation 12: dropping the lower-order terms

Given two asymptotically non-negative functions f and g of a natural variable, we have

$$g(n) = \mathcal{O}\big(f(n)\big) \quad \Longrightarrow \quad f(n) + g(n) = \Theta\big(f(n)\big).$$

Proof.

- Let $0 \leq g(n) \leq c\, f(n)$ for all $n \geq n_1'$, and let $f(n) \geq 0$ for all $n \geq n_1''$. Then, for all $n \geq n_1 = \max\{n_1', n_1''\}$,

$$0 \leq 1 \cdot f(n) \leq f(n) + g(n) \leq f(n) + c\, f(n) = (1 + c)\, f(n),$$

that is, $f(n) + g(n) = \Theta\big(f(n)\big)$ with the constants $c_1 = 1$ and $c_2 = 1 + c$

□

We now apply the notation
to the number of steps of two algorithms
of the chapter «Running time».

Example: the number of steps of two algorithms (1/4)

In the chapter «Running time» we count the steps of two algorithms on an array of n entries.

- `FindMaximum` executes $S(n) = 4n$ steps
- By choosing $c_1 = c_2 = 4$ and $n_1 = 1$ we have

$$0 \leq 4\,n \leq S(n) \leq 4\,n, \qquad \forall n \geq 1,$$

so $S(n) = \Theta(n)$: the number of steps is **linear** in n

Example: the number of steps of two algorithms (2/4)

- `InsertionSort` executes, in the worst case,

$$S(n) = \frac{3n^2 + 9n - 8}{2}$$

steps. To prove that $S(n) = \Theta(n^2)$ we determine $c_1, c_2 \in \mathbb{R}_{>0}$ and $n_1 \in \mathbb{N}$ such that

$$0 \leq c_1\, n^2 \leq \frac{3n^2 + 9n - 8}{2} \leq c_2\, n^2, \qquad \forall n \geq n_1.$$

Example: the number of steps of two algorithms (3/4)

- For the lower bound, $9n - 8 \geq 0$ for all $n \geq 1$, hence $\frac{3n^2+9n-8}{2} \geq \frac{3}{2}\, n^2$
- For the upper bound, $9n - 8 \leq 9n \leq 9n^2$ for all $n \geq 1$, hence $\frac{3n^2+9n-8}{2} \leq \frac{12\, n^2}{2} = 6\, n^2$

- By choosing $c_1 = \frac{3}{2}$, $c_2 = 6$ and $n_1 = 1$, we have

$$0 \leq \frac{3}{2}\, n^2 \leq \frac{3n^2 + 9n - 8}{2} \leq 6\, n^2, \qquad \forall n \geq 1.$$

Example: the number of steps of two algorithms (4/4)

- The same result follows from the limit $\lim_{n\to+\infty} \frac{3n^2+9n-8}{2n^2} = \frac{3}{2} \in \mathbb{R}_{>0}$ and (11). The number of steps is **quadratic** in n
- Since $n = o(n^2)$, for large arrays `InsertionSort` executes, in the worst case, many more steps than `FindMaximum`: by Observation 5, the ratio $\frac{4n}{(3n^2+9n-8)/2}$ tends to 0^+

$\mathcal{O}$, Ω and Θ for the number of steps

- The three notations $\mathcal{O}$, Ω and Θ say different things about the number of steps $S(n)$ in the worst case
- If $S(n) = \mathcal{O}\big(g(n)\big)$, then, for all $n \geq n_1$, the algorithm executes at most $c\,g(n)$ steps on **every** input with parameter n
- If $S(n) = \Omega\big(g(n)\big)$, then, for all $n \geq n_1$, there is **at least one** input with parameter n on which the algorithm executes at least $c\,g(n)$ steps
- The statement $S(n) = \Theta\big(g(n)\big)$ gives both, and it is the most informative one

- In the chapter «Running time» the order of growth of $S(n)$ is then expressed as a function of the input size $|I|$, using the order of growth of $|I|$ obtained in the chapter «Input size»

Summary

- The asymptotic notation keeps the **order of growth** of the number of steps, which does not depend on the counting conventions or on the computer
- Constant factors and lower-order terms do not matter: if $g(n) = \mathcal{O}\big(f(n)\big)$, then $f(n) + g(n) = \Theta\big(f(n)\big)$
- `FindMaximum` executes $\Theta(n)$ steps, `InsertionSort` $\Theta(n^2)$ steps in the worst case
- $\mathcal{O}$ bounds the steps on every input, Ω on at least one input, Θ gives both

Summary (1/3)

- $f(n) = \Theta\big(g(n)\big)$, $\mathcal{O}\big(g(n)\big)$, $\Omega\big(g(n)\big)$: there exist positive constants (c_1 and c_2, or c) and a threshold n_1 such that, for all $n \geq n_1$, $f(n)$ lies between $c_1 g(n)$ and $c_2 g(n)$, below $c\, g(n)$, above $c\, g(n)$. To prove these statements we determine the constants

- $f(n) = o\big(g(n)\big)$, $\omega\big(g(n)\big)$: the inequality $f(n) < c\, g(n)$, or $c\, g(n) < f(n)$, holds for **every** $c > 0$ from a threshold n_1 that depends on c

- Θ is the same as $\mathcal{O}$ and Ω together; o implies $\mathcal{O}$ and ω implies Ω. None of these implications can be reversed

Summary (2/3)

- **Limit criterion**: for asymptotically positive functions whose ratio $\frac{f(n)}{g(n)}$ has a limit, the limit decides:
  - 0^+ gives o and $\mathcal{O}$
  - a number $\ell > 0$ gives Θ, $\mathcal{O}$ and Ω
  - $+\infty$ gives ω and Ω

  and in each case the other notations do not hold

- When the limit does not exist, as for $2 - \sin n$ or $2 + (-1)^n$, we use the definitions

Summary (3/3)

- The notations are transitive; Θ, $\mathcal{O}$, Ω are reflexive; Θ is symmetric; $\mathcal{O}$ and Ω, and o and ω, are transpose symmetric
- Unlike real numbers, two functions may not be comparable at all, as $n^{1+\sin n}$ and n
- In the analysis of algorithms, the notation drops constant factors and lower-order terms and keeps the order of growth of the number of steps